

PRACTICE SHEET

1. If $u = \sin^{-1}(x - y)$, $x = 3t$, $y = t^3$, then what is the derivative of u with respect to ?
 (a) $3(1 - t^2)$ (b) $3(1 - t^2)^{-\frac{1}{2}}$
 (c) $5(1 - t^2)^{-\frac{1}{2}}$ (d) $5(1 - t^2)$
2. If $y = x + e^x$, then what is $\frac{d^2x}{dy^2}$ equal to?
 (a) e^x (b) $-\frac{e^x}{(1 + e^x)^3}$
 (c) $-\frac{e^x}{(1 + e^x)}$ (d) $-\frac{e^x}{(1 + e^x)^2}$
3. If, $x + y = t - \frac{1}{t}$, $x^2 + y^2 = t^2 + \frac{1}{t^2}$ What is $\frac{dy}{dx}$ equal to?
 (a) $\frac{1}{x}$ (b) $-\frac{1}{x}$
 (c) $\frac{1}{x^2}$ (d) $-\frac{1}{x^2}$
4. What is the derivative of $\cos^{-1}\left(\frac{2\cos x + 3\sin x}{\sqrt{13}}\right)$?
 (a) $\frac{1}{\sqrt{1-x^2}}$ (b) $-\frac{1}{\sqrt{1-x^2}}$
 (c) 0 (d) 1
5. What is the solution of $y' = 1 + x + y^2 + xy^2$, $y(0) = 0$?
 (a) $y = \tan^2\left(\frac{x^2}{2} + x\right)$ (b) $y = \tan^2(x^2 + x)$
 (c) $y = \tan(x^2 + x)$ (d) $y = \tan\left(\frac{x^2}{2} + x\right)$
6. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, what is $\frac{dy}{dx}$ equal to?
 (a) $-\frac{1}{1+x}$ (b) $-\frac{1}{(1+x)^2}$
 (c) $\frac{1}{(1+x)^2}$ (d) $\frac{\sqrt{x}}{\sqrt{1+x}}$
7. If $x = \sin t - t \cos t$ and $y = t \sin t + \cos t$, then what is $\frac{dy}{dx}$ at point $t = \frac{\pi}{2}$?
 (a) 0 (b) $\frac{\pi}{2}$
 (c) $-\frac{\pi}{2}$ (d) 1
8. If $y = \sin^{-1}x + \sin^{-1}\sqrt{1-x^2}$ what is $\frac{dy}{dx}$ equal to?
 (a) $\cos^{-1}x + \cos^{-1}\sqrt{1-x^2}$ (b) $\frac{1}{\cos x} + \frac{1}{\cos\sqrt{1-x^2}}$
 (c) $\frac{\pi}{2}$ (d) 0
9. If $f(x) = \log_e[\log_e x]$, then what is $f'(e)$ equal to?
 (a) e^{-1} (b) e
 (c) 1 (d) 0
10. For the curve $\sqrt{x} + \sqrt{y} = 1$, what is the value of $\frac{dy}{dx}$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$?
 (a) $\frac{1}{2}$ (b) 1
 (c) -1 (d) 2
11. If $y = \frac{1}{\log_{10} x}$, then what is $\frac{dy}{dx}$ equal to
 (a) x (b) $x \log_e 10$
 (c) $-\frac{(\log_x 10)^2 (\log 10^e)}{x}$ (d) $x \log_{10} e$
12. If $x^y = e^{x-y}$, then dy/dx is equal to which one of the following?
 (a) $\frac{(x-y)}{(1+\log x)^2}$ (b) $\frac{y}{(1+\log x)}$
 (c) $\frac{(x+y)}{(1+\log x)}$ (d) $\frac{(\log x)}{(1+\log x)^2}$
13. If $f(x) = \cos x$, $g(x) = \log x$, and $y = (g \circ f)(x)$, then what is the value of $\frac{dy}{dx}$ at $x = 0$?
 (a) 0 (b) 1
 (c) -1 (d) 2
14. What is the derivative of $\log_x 5$ respect to $\log_5 x$?
 (a) $-(\log_5 x)^{-2}$ (b) $(\log_5 x)^{-2}$
 (c) $-(\log_x 5)^{-2}$ (d) $(\log_x 5)^{-2}$
15. If $f(x) = \sin^2 x^2$, then what is $f'(x)$ equal to?
 (a) $4x \sin(x^2) \cos(x^2)$ (b) $2 \sin(x^2) \cos(x^2)$
 (c) $4 \sin(x^2) \sin^2 x$ (d) $2x \cos^2(x^2)$
16. If $f(x) = \tan x + e^{-2x} - 7x^3$, then what is the value of $f'(0)$?
 (a) -2 (b) -1

- (c) 0 (d) 3
17. If $3^x + 3^y = 3^{x+y}$, then what is $\frac{dy}{dx}$ equal to?
- (a) $\frac{3^{x+y} - 3^x}{3^y}$ (b) $\frac{3^{x-y}(3^y - 1)}{1 - 3^x}$
- (c) $\frac{3^x + 3^y}{3^x - 3^y}$ (d) $\frac{3^x + 3^y}{1 + 3^{x+y}}$
18. If $y = \sin(m \sin^{-1} x)$, then what is the value of d^2y/dx^2 at $x = 0$?
- (a) m (b) m^2
- (c) $m^2 + 2$ (d) 0
19. If $y = f(x)$, $p = \frac{dy}{dx}$ and $q = \frac{d^2y}{dx^2}$, then what is $\frac{d^2x}{dy^2}$ equal to?
- (a) $-\frac{q}{p^2}$ (b) $-\frac{q}{p^3}$
- (c) $\frac{1}{q}$ (d) $\frac{q}{p^2}$
20. If $f(x) = e^{\sin(\log \cos x)}$ and $g(x) = \log \cos x$, then what is the derivative of $f(x)g(x)$?
- (a) $f(x)\cos[g(x)]$ (b) $f(x)\sin[g(x)]$
- (c) $g(x)\cos[f(x)]$ (d) $g(x)\sin[f(x)]$
21. What is the derivative of $\sin^2 x$ with respect to $\cos^2 x$?
- (a) $\tan^2 x$ (b) $\cot^2 x$
- (c) -1 (d) 1
22. If $\sqrt{x} + \sqrt{y} = 2$, then what is $\frac{dy}{dx}$ at $y = 1$ equal to?
- (a) 5 (b) 4
- (c) 2 (d) -1
23. If $x = k(\theta + \sin \theta)$ and $y = k(1 + \cos \theta)$, then what is the derivative of y with respect to x at $\theta = \pi/2$?
- (a) -1 (b) 0
- (c) 1 (d) 2
24. If $x = t^2, y = t^3$, what is $\frac{d^2y}{dx^2}$ equal to?
- (a) 1 (b) $\frac{3}{2t}$
- (c) $\frac{3}{4t}$ (d) $\frac{3}{2}$
25. If $y = \sin^{-1}\left(\frac{4x}{1+4x^2}\right)$ then what is $\frac{dy}{dx}$ equal to?
- (a) $\frac{1}{1+4x^2}$ (b) $-\frac{1}{1+4x^2}$
- (c) $\frac{4}{1+4x^2}$ (d) $\frac{4x}{1+4x^2}$
26. What is the derivative of $x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right)$?
- (a) $\sqrt{a^2 - x^2}$ (b) $2\sqrt{a^2 - x^2}$
- (c) $\sqrt{x^2 - a^2}$ (d) $2\sqrt{x^2 - a^2}$

ANSWER KEYS

1.	b	2.	b	3.	c	4.	d	5.	d	6.	b	7.	a	8.	d	9.	a	10.	c
11.	c	12.	d	13.	a	14.	a	15.	a	16.	b	17.	b	18.	d	19.	b	20.	a
21.	c	22.	d	23.	a	24.	c	25.	c	26.	b								

Solutions

Sol. 1. (b)

$$u = \sin^{-1}(x - y) \text{ and } x = 3t, y = 4t^3$$

$$\text{So, } u = \sin^{-1}(3t - 4t^3)$$

$$\text{Let } t = \sin \theta \Rightarrow \theta = \sin^{-1} t,$$

$$\text{So, } u = \sin^{-1}(3 \sin \theta - 4 \sin^3 \theta)$$

$$= \sin^{-1}(\sin 3\theta) = 3\theta$$

$$\text{Hence, } u = 3 \sin^{-1} t$$

$$\frac{du}{dt} = 3 \frac{1}{\sqrt{1-t^2}} = 3(1-t^2)^{-1/2}$$

Sol. 2. (b)

$$\text{Given that } y = x + e^x$$

Differentiating w.r.t. x

$$\frac{dy}{dx} = 1 + e^x$$

$$\Rightarrow \frac{dx}{dy} = \frac{1}{1+e^x}$$

Differentiating w.r.t.

$$\frac{d^2x}{dy^2} = -\frac{(1)(e^x)}{(1+e^x)^2} \cdot \frac{dx}{dy}$$

$$= -\frac{e^x}{(1+e^x)} \cdot \frac{1}{(1+e^x)} = -\frac{e^x}{(1+e^x)^2}$$

Sol. 3. (c)

$$\text{Given that } x + y = t - 1/t \text{ and}$$

$$x^2 + y^2 = t^2 + \frac{1}{t^2}$$

$$\therefore (x+y)^2 = x^2 + y^2 + 2xy$$

$$\Rightarrow \left(t - \frac{1}{t}\right)^2 + \left(t^2 + \frac{1}{t^2}\right) + 2xy$$

$$-2 = 2xy \Rightarrow xy = -1$$

$$(x-y)^2 = (x+y)^2 - 4xy$$

$$= \left(t - \frac{1}{t}\right)^2 - 4 \times -1 = t^2 + \frac{1}{t^2} - 2 + 4 = \left(t - \frac{1}{t}\right)^2$$

$$x - y = t + 1/t$$

$$\Rightarrow x = t, y = -1/t$$

$$xy = -1$$

$$\Rightarrow x \frac{dy}{dx} + y = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x} = \frac{1}{t^2} = \frac{1}{x^2}$$

Sol. 4. (d)

The given function

$$f(x) = \cos^{-1} \left(\frac{2 \cos x + 3 \sin x}{\sqrt{13}} \right)$$

Can be written as

$$\cos^{-1} \left(\cos x \cdot \frac{2}{\sqrt{13}} + \sin x \cdot \frac{3}{\sqrt{13}} \right)$$

$$\text{let } \frac{2}{\sqrt{13}} = \cos \theta \text{ and } \frac{3}{\sqrt{13}} = \sin \theta$$

$$\Rightarrow \tan \theta = \frac{\frac{3}{\sqrt{13}}}{\frac{2}{\sqrt{13}}} = \frac{3}{2}$$

$$\text{So, (i), } \cos^{-1}(\cos x \cos \theta + \sin x \sin \theta)$$

$$\text{Where } \theta = \tan^{-1} \left(\frac{3}{2} \right)$$

$$= \cos^{-1}(\cos(x-\theta)) = x - \theta$$

$$\text{Hence, } f'(x) = 1 \quad (\because \theta \text{ is constant})$$

Sol. 5. (d)

Given differential equation is

$$\frac{dy}{dx} = 1 + x + y^2 + xy^2$$

$$\Rightarrow \frac{dy}{dx} = (1+x)(1+y^2)$$

$$\Rightarrow \frac{dy}{1+y^2} = (1+x)dx$$

$$\Rightarrow \tan^{-1} y = \frac{x^2}{2} + x + c$$

$$\text{Given that when } x = 0, y(0) = 0.$$

$$\text{Hence, } c = 0$$

$$\Rightarrow y = \tan \left(\frac{x^2}{2} + x \right)$$

Sol. 6. (b)

Given equation

$$x\sqrt{1+y} + y\sqrt{1+x} = 0$$

Can be written as:

$$x\sqrt{1+y} = -y\sqrt{1+x}$$

Squaring both sides, we get

$$x^2(1+y) = y^2(1+x)$$

$$\Rightarrow x^2 + x^2y = y^2 + y^2x$$

$$\Rightarrow x^2 - y^2 = y^2x - x^2y$$

$$\Rightarrow (x-y)(x+y) = -xy(x-y)$$

$$\Rightarrow x+y = -xy \Rightarrow y(1+x) = -x$$

$$y = \frac{-x}{1+x} \text{ which is in explicit form}$$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{(1+x)(-1) + x(1)}{(1+x)^2} = \frac{-1}{(1+x)^2}$$

Sol. 7. (a)

As given

$$x = \sin t - t \cos t \text{ and } y = t \sin t + \cos t$$

On differentiating w.r.t. x, we get

$$\frac{dx}{dt} = \cos t - \{\cos t + t(-\sin t)\}$$

$$\Rightarrow \frac{dx}{dt} = \cos t - \cos t + t \sin t = t \sin t$$

$$\text{and } \frac{dy}{dt} = t \cos t + \sin t - \sin t = t \cos t$$

$$\text{Hence, } = \cot t$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{t=\frac{\pi}{2}} = \cot \frac{\pi}{2} = 0$$

Sol. 8. (d)

Given function is:

$$y = \sin^{-1} x + \sin^{-1} \sqrt{1-x^2}$$

on differentiating, w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-1+x^2}} \cdot \frac{1}{2\sqrt{1-x^2}} (-2x)$$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$$

Sol. 9. (a)

The given function is: $f(x) = \log_e [\log_e x]$

Differentiating w.r.t. x, we get

$$f'(x) = \frac{1}{\log_e x} \cdot \frac{1}{x} \Rightarrow f'(e) = \frac{1}{\log_e e} \cdot \frac{1}{e} = \frac{1}{e} = e^{-1}$$

Sol. 10. (c)

$$\text{Given function: } \sqrt{x} + \sqrt{y} = 1$$

is an implicit function

Differentiating both sides w.r.t. x, we get

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\sqrt{\frac{y}{x}}$$

$$\text{Value of } \frac{dy}{dx} \text{ at } x = \frac{1}{4}, y = \frac{1}{4}$$

$$\left(\frac{dy}{dx} \right)_{\left(\frac{1}{4}, \frac{1}{4} \right)} = -\sqrt{\frac{1/4}{1/4}} = -1$$

Sol. 11. (c)

$$\text{Differentiating the given function, } y = \frac{1}{\log_{10} x}$$

$$\text{We get, } \frac{dy}{dx} = -\frac{1}{(\log_{10} x)^2} \cdot \frac{1}{x} \log_{10} e$$

$$\Rightarrow \frac{dy}{dx} = -\frac{(\log_{10} 10)^2 \cdot \log_{10} e}{x}$$

Sol. 12. (d)

$$x^y = e^{x^y}$$

Taking log both sides, we get

$$\Rightarrow y, \log x = x - y$$

$$\Rightarrow y = \frac{x}{1 + \log x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1 + \log x) - x \left(0 + \frac{1}{x} \right)}{(1 + \log x)^2}$$

$$= \frac{(1 + \log x) - 1}{(1 + \log x)^2} = \frac{\log x}{(1 + \log x)^2}$$

Sol. 13. (a)

$$\therefore y = \log f(x) = g\{f(x)\} \log(\cos x)$$

$$\therefore \frac{dy}{dx} = \frac{1}{\cos x} (-\sin x) = -\tan x$$

$$\therefore \left(\frac{dy}{dx} \right)_{\text{at } x=0} = -\tan 0 = 0$$

Sol. 14. (a)

$$\text{Let } u = \log_x 5, \text{ and } v = \log_5 x$$

$$\Rightarrow u = \frac{\log_e 5}{\log_e x}, \text{ and } v = \frac{\log_e x}{\log_e 5}$$

On differentiating w. r. t. x., we get

$$\frac{du}{dx} = -\frac{\log_e 5}{x(\log_e x)^2}, \text{ and } \frac{dv}{dx} = \frac{1}{x \log_e 5}$$

$$\therefore \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\log_e 5}{x(\log_e x)^2} \times x \log_e = -\left(\frac{\log_e 5}{\log_e x}\right)^2$$

$$= -(\log_x 5)^2 = -(\log_5 x)^{-2}$$

Sol. 15. (a)

$$f(x) = \sin^2 x^2,$$

$$f'(x) = 4x \sin(x^2) \cos(x^2)$$

Sol. 16. (b)

$$f(x) = \tan x + e^{-2x} - 7x^3,$$

$$f'(x) = \sec^2 x - 2e^{-2x} - 21x^2$$

$$f'(0) = 1 - 2 = -1$$

Sol. 17. (b)

$$3^x + 3^y = 3^{x+y}$$

On differentiating w. r. t. x., we get

$$3^x \log 3 + 3^y \log 3 \frac{dy}{dx} = 3^{(x+y)} \log 3 \left(1 + \frac{dy}{dx}\right)$$

$$\Rightarrow 3^x + 3^y \frac{dy}{dx} = 3^{x+y} + 3^{(x+y)} \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} (-3^{x+y} + 3^y) = 3^{3+y} - 3^x$$

$$\Rightarrow \frac{dy}{dx} = \frac{3^x (3^y - 1)}{3^y (1 - 3^x)} = \frac{3^{x-y} (x^y - 1)}{(1 - 3^x)}$$

Sol. 18. (d)

$$y = \sin(m \sin^{-1} x),$$

$$\text{put } m = 2$$

$$y = \sin(2 \sin^{-1} x)$$

$$y = \sin(\sin^{-1} 2x \sqrt{1-x^2}) = 2x \sqrt{1-x^2}$$

$$\frac{dy}{dx} = 2\sqrt{1-x^2} + \frac{2x(-2x)}{2\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{2(1-x^2) - 2x^2}{\sqrt{1-x^2}} = \frac{2-4x^2}{\sqrt{1-x^2}}$$

$$\frac{d^2y}{dx^2} = \frac{\sqrt{1-x^2}(-8x) - (2-4x^2) \frac{-2x}{2\sqrt{1-x^2}}}{(1-x^2)}$$

$$\text{put } x = 0$$

$$\frac{d^2y}{dx^2} = 0$$

Sol. 19. (b)

$$\text{We have, } \frac{dy}{dx} = p$$

$$\frac{dx}{dy} = \frac{1}{p} \Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{p^2} \frac{dp}{dy}$$

$$\Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{p^2} \frac{dp}{dy} = -\frac{1}{p^2} \frac{dp}{dx} \frac{dx}{dy} = -\frac{q}{p^3}$$

Sol. 20. (a)

$$\therefore f(x) = e^{\sin(\log \cos x)}$$

$$\therefore f'(x) = e^{\sin(\log \cos x)} \cdot \cos(\log \cos x) \cdot \frac{1}{\cos x} (-\sin x)$$

$$= -e^{\sin(\log \cos x)} \cdot \cos(\log \cos x) \cdot \tan x$$

$$\text{and } g(x) = \log \cos x$$

$$\therefore g'(x) = \frac{1}{\cos x} (-\sin x) = -\tan x$$

$$\text{Hence, } \frac{f'(x)}{g'(x)} = \frac{-e^{\sin(\log \cos x)} \cdot \cos(\log \cos x) \cdot \tan x}{-\tan x}$$

$$= e^{\sin(\log \cos x)} \cdot \cos(\log \cos x)$$

$$= f(x) \cdot \cos[g(x)]$$

Sol. 21. (c)

$$\text{Let } u = \sin^2 x, \text{ and } v = \cos^2 x$$

$$\Rightarrow \frac{du}{dx} = 2 \sin x \cos x$$

$$\text{and } \frac{dv}{dx} = -2 \sin x \cos x$$

$$\therefore \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{2 \sin x \cos x}{-2 \sin x \cos x} = -1$$

Sol. 22. (d)

$$\sqrt{x} + \sqrt{y} = 2,$$

$$\text{at } y = 1, x = 1$$

On differentiating w. r. t. x., we get

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0,$$

$$\frac{dy}{dx} = -\sqrt{\frac{y}{x}} = -1$$

Sol. 23. (a)

$$\therefore x = k(\theta + \sin \theta) \text{ and } y = k(1 + \cos \theta)$$

$$\Rightarrow \frac{dy}{d\theta} = k(1 + \cos \theta) \text{ and } \frac{dy}{dx} = -k \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{-k \sin \theta}{k(1 + \cos \theta)} = \frac{-2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = -\tan \frac{\theta}{2}$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{2}} = -\tan \frac{\pi}{4} = -1$$

Sol. 24. (c)

$$x = t^2, y = t^3,$$

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2, \frac{dy}{dx} = \frac{3}{2}t,$$

$$\frac{d^2y}{dx^2} = \frac{3}{2} \frac{dt}{dx} = \frac{3}{4t}$$

Sol. 25. (c)

$$y = \sin^{-1} \left(\frac{4x}{1+4x^2} \right)$$

$$\text{put } 2x = \tan t$$

$$y = \sin^{-1}(\sin 2t) = 2t = 2 \tan^{-1}(2x)$$

$$\frac{dy}{dx} = \frac{4}{1+4x^2},$$

Sol. 26. (b)

$$\frac{dy}{dx} = x \cdot \frac{1}{2\sqrt{a^2-x^2}} \cdot (-2x) + \sqrt{a^2-x^2} + \frac{a^2}{\sqrt{1-\frac{x^2}{a^2}}} \cdot \frac{1}{a}$$

$$= \frac{-x^2}{\sqrt{a^2-x^2}} + \sqrt{a^2-x^2} + \frac{a^2 \cdot a}{\sqrt{a^2-x^2}} \cdot \frac{1}{a}$$

$$= \frac{a^2-x^2}{\sqrt{a^2-x^2}} + \sqrt{a^2-x^2} = 2\sqrt{a^2-x^2}$$

NDA PYQ

1. If $f(x) = 2^x$, then what is the double derivative of $f''(x)$?
 (a) $2^x (\ln 2)^2$ (b) $x(x-1)2^{x-2}$
 (c) $2^{x+2} (\ln 2)$ (d) $2^x (\ln_{10} 2)^2$
[NDA (I) - 2011]
2. If $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, then what is $\frac{dy}{dx}$ equal to?
 (a) 1 (b) -1
 (c) 0 (d) -2x
[NDA (II) - 2011]
3. If $y = \ln \sqrt{\tan x}$, then what is the value of $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$?
 (a) 0 (b) -1
 (c) $1/2$ (d) 1
[NDA-2011(2)]
4. If $y = \cos t$ and $x = \sin t$, then what is the value of $\frac{dy}{dx}$?
 (a) xy (b) x/y
 (c) $-y/x$ (d) $-x/y$
[NDA (I) - 2012]
5. If $y = \frac{x+1}{x-1}$, then what is the value of $\frac{dy}{dx}$?
 (a) $\frac{-2}{x-1}$ (b) $\frac{-2}{(x-1)^2}$
 (c) $\frac{2}{(x-1)^2}$ (d) $\frac{2}{(x-1)}$
[NDA (I) - 2012]
6. What is the rate of change of $\sqrt{x^2+16}$ w.r.t. x^2 at $x = 3$?
 (a) $\frac{1}{5}$ (b) $\frac{1}{10}$
 (c) $\frac{1}{20}$ (d) $\frac{1}{15}$
[NDA (I) - 2012]
7. If $f(x) = x^2 - 6x + 8$ and there exists a point c in the interval $[2, 4]$ such that $f'(c) = 0$, then what is the value of c ?
 (a) 2.5 (b) 2.8
 (c) 3 (d) 3.5
[NDA-2012(1)]
8. If $f(x) = 2^{\sin x}$, then what is the derivative of $f'(x)$?
 (a) $2^{\sin x} \ln 2$ (b) $(\sin x)2^{\sin x-1}$
 (c) $(\cos x)2^{\sin x-1}$ (d) None of these
[NDA (II) - 2012]
9. If $x^m + y^m = 1$, such that $\frac{dy}{dx} = -\frac{x}{y}$, then what should be the value of m ?
 (a) 0 (b) 1
 (c) 2 (d) -1
[NDA (II) - 2012]
10. If $y = \ln(e^{mx} + e^{-mx})$, then what is the value of $\frac{dy}{dx}$ at $x = 0$?
 (a) -1 (b) 0
 (c) 1 (d) 2
[NDA (II) - 2012]
11. Consider the following statements:
 I. If $y = \ln(\sec x + \tan x)$, then $\frac{dy}{dx} = \sec x$
 II. If $y = \ln(\operatorname{cosec} x - \cot x)$, then $\frac{dy}{dx} = \operatorname{cosec} x$
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
[NDA (II) - 2012]
12. If $2x^3 - 3y^2 = 7$, then what is $\frac{dy}{dx}$ equal to (where, $y \neq 0$)?
 (a) $\frac{x^2}{2y}$ (b) $\frac{x}{2y}$
 (c) $\frac{x^2}{y}$ (d) None of these
[NDA (I) - 2013]
13. The derivation of $\sec^2 x$ w.r.t $\tan^2 x$ is
 (a) 1 (b) 2
 (c) $2 \sec x \tan x$ (d) $2 \sec^2 x \tan x$
[NDA (I) - 2013]
14. What is the differentiation on $\log_x x$?
 (a) 0 (b) 1
 (c) $1/x$ (d) x
[NDA (I) - 2013]
15. If $y = x^x$, then what is the value of $\frac{dy}{dx}$ at $x = 1$?
 (a) 0 (b) 1
 (c) -1 (d) 2
[NDA (I) - 2013]
16. If $y = \sin(ax+b)$, then what is $\frac{d^2y}{dx^2}$ at $x = -\frac{b}{a}$, where a and b are constants and $a \neq 0$?
 (a) 0 (b) -1
 (c) $\sin(a-b)$ (d) $\sin(a+b)$
[NDA (I) - 2013]
17. What is the derivation of x^3 w.r.t. x^2 ?
 (a) $3x^2$ (b) $\frac{3x}{2}$
 (c) x (d) $\frac{3}{2}$
[NDA (II) - 2013]

18. What is the derivation of $\sin(\sin x)$?

- (a) $\cos(\cos x)$ (b) $\cos(\sin x)$
(c) $\cos(\sin x)\cos x$ (d) $\cos(\cos x)\cos x$

[NDA (II) - 2013]

19. What is the derivation of $|x-1|$ at $x=2$?

- (a) -1 (b) 0
(c) 1 (d) does not exist

[NDA (II) - 2013]

20. If $f(x) = 2x^2 + 3x - 5$, then what is $f'(0) + 3f'(-1)$ =

- (a) -1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2013]

21. What is the derivative of $\sqrt{\frac{1+\cos x}{1-\cos x}}$

- (a) $\frac{1}{2}\sec^2 \frac{x}{2}$
(b) $-\frac{1}{2}\operatorname{cosec}^2 \frac{x}{2}$
(c) $-\operatorname{cosec}^2 \frac{x}{2}$
(d) None of the above

[NDA (I) - 2014]

22. If $z = f \circ f(x)$ where $f(x) = x^2$, then what is $\frac{dz}{dx}$ equal to?

- (a) x^3 (b) $2x^3$
(c) $4x^3$ (d) $4x^2$

[NDA-2014(1)]

Direction (for next two): Consider the curve

$$x = a(\cos \theta + \theta \sin \theta) \text{ and } y = a(\sin \theta - \theta \cos \theta).$$

23. What is $\frac{dy}{dx}$ equal to?

- (a) $\tan \theta$ (b) $\cot \theta$
(c) $\sin 2\theta$ (d) $\cos 2\theta$

[NDA (II) - 2014]

24. What is $\frac{d^2y}{dx^2}$ equal to?

- (a) $\sec^2 \theta$ (b) $-\operatorname{cosec}^2 \theta$
(c) $\frac{\sec^3 \theta}{a\theta}$ (d) None of these

[NDA (II) - 2014]

25. If $y = x \ln x + xe^x$, then what is the value of $\frac{dy}{dx}$ at $x = 1$?

- (a) $1+e$ (b) $1-e$
(c) $1+2e$ (d) None of these

[NDA (II) - 2014]

For the next three (03) items that follow:

Let $f(x) = ax^2 + bx + c$ such that $f(1) = f(-1)$ and a, b, c are in Arithmetic progression.

26. What is the value of b ?

- (a) -1
(b) 0
(c) 1
(d) cannot be determined due to insufficient data

27. $f(a), f(b), f(c)$ are in

- (a) A.P.
(b) G.P.
(c) H.P.
(d) Arithmetic co-geometric progression

[NDA-2014(2)]

28. $f'(a), f'(b), f'(c)$ are:

- (a) in A.P. only
(b) in G.P. only
(c) in both A.P. and G.P.
(d) neither in A.P. nor in G.P.

[NDA-2014(2)]

[NDA-2014(2)]

Direction (for next two): Given that,

$$\frac{d}{dx} \left(\frac{1+x^2+x^4}{1+x+x^2} \right) = Ax + B$$

29. What is the value of A ?

- (a) -1 (b) 1
(c) 2 (d) 4

[NDA (I) - 2015]

30. What is the value of B ?

- (a) -1 (b) 1
(c) 2 (d) 4

[NDA (I) - 2015]

31. What is the derivative of $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} x$?

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) x

[NDA (I) - 2015]

Direction (for next three)

Consider the parametric equation

$$x = \frac{a(1-t^2)}{1+t^2} \text{ and } y = \frac{2at}{1+t^2}$$

32. What does the equation represent?

- (a) It represents a circle of diameter a
(b) It represents a circle of radius a
(c) It represents a parabola
(d) None of the above

[NDA (I) - 2015]

33. What is $\frac{dy}{dx}$ equal to?

- (a) y/x (b) $-y/x$
(c) x/y (d) $-x/y$

[NDA (I) - 2015]

34. What is $\frac{d^2y}{dx^2}$ equal to

- (a) $\frac{a^2}{y^2}$ (b) $\frac{a^2}{x^2}$
(c) $-\frac{a^2}{x^2}$ (d) $-\frac{a^2}{y^2}$

[NDA (I) - 2015]

35. If $y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$, where $0 < x < \frac{\pi}{2}$,

then $\frac{dy}{dx}$ is equal to

- (a) $\frac{1}{2}$ (b) 2
(c) $\sin x + \cos x$ (d) $\sin x - \cos x$

[NDA (II) - 2015]

36. If $s = \sqrt{t^2 + 1}$, then $\frac{d^2s}{dt^2}$ is equal to

- (a) $\frac{1}{s}$ (b) $\frac{1}{s^2}$
(c) $\frac{1}{s^3}$ (d) $\frac{1}{s^4}$

[NDA (II) - 2015]

37. The derivative of $\ln(x + \sin x)$ with respect to $(x + \cos x)$ is

- (a) $\frac{1 + \cos x}{(x + \sin x)(1 - \sin x)}$ (b) $\frac{1 - \cos x}{(x + \sin x)(1 + \sin x)}$
(c) $\frac{1 - \cos x}{(x - \sin x)(1 + \cos x)}$ (d) $\frac{1 + \cos x}{(x - \sin x)(1 - \cos x)}$

[NDA (II) - 2015]

38. If $x^a y^b = (x - y)^{a+b}$, then the value of $\frac{dy}{dx} - \frac{y}{x}$ is equal to:

- (a) a/b (b) b/a
(c) 1 (d) 0

[NDA (II)-2015]

For the next two (2) items that follow:

Consider the function

$$f(x) = |x^2 - 5x + 6|$$

39. What is $f'(4)$ equal to?

- (a) -4 (b) -3
(c) 3 (d) 2

[NDA (I) - 2016]

40. What is $f''(2.5)$ equal to?

- (a) -3 (b) -2
(c) 0 (d) 2

[NDA (I) - 2016]

Direction (for next two):

Consider the equation $x + |y| = 2y$

41. Which of the following statements are not correct?

1. y as a function of x is not defined for all real x .
2. y as a function of x is not continuous at $x = 0$
3. y as a function of x is differentiable for all x .

Select the correct answer using the code given below:

- (a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

[NDA (I) - 2016]

42. What is the derivative of y as a function of x with respect to x for $x < 0$?

- (a) 2 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

[NDA (I) - 2016]

For the next two (2) items that follow:

Consider the function

$$f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix} \text{ where } p \text{ is a constant.}$$

43. What is the value of $f'(0)$?

- (a) p^3 (b) $3p^3$
(c) $6p^3$ (d) $-6p^3$

[NDA (I) - 2016]

44. What is the value of p for which $f''(0) = 0$?

- (a) $-\frac{1}{6}$ or 0 (b) -1 or 0
(c) $\frac{1}{6}$ or 0 (d) -1 or 1

[NDA (I) - 2016]

For the next four (4) items that follow:

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for $x \in \mathbb{R}$

45. What is $f(1)$ equal to:

- (a) -2 (b) -1
(c) 0 (d) 4

[NDA (I) - 2016]

46. What is $f'(1)$ is equal to ?

- (a) -5 (b) -1
(c) 1 (d) 0

[NDA (I) - 2016]

47. What is $f'''(10)$ equal to?

- (a) 1 (b) 5
(c) 6 (d) 0

[NDA (I) - 2016]

48. Consider the following:

$$1. f(2) = f(1) - f(0)$$

$$2. f''(2) - 2f'(1) = 12$$

Which of the above is / are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2016]

49. If $y = \log_{10} x + \log_x 10 + \log_x x + \log_{10} 10$ then what is

$$\left(\frac{dy}{dx}\right)_{x=10} \text{ equal to ?}$$

- (a) 10 (b) 2
(c) 1 (d) 0

[NDA (I) - 2016]

Direction (for next three): Consider the following for the next two items that follow:

$$\text{Let } f(x) = [|x| - |x - 1|]^2$$

50. What is $f'(x)$ equal to when $x > 1$?

- (a) 0 (b) $2x - 1$
(c) $4x - 2$ (d) $8x - 4$

[NDA (II) - 2016]

51. What is $f'(x)$ equal to when $0 < x < 1$?

- (a) 0 (b) $2x - 1$
(c) $4x - 2$ (d) $8x - 4$

[NDA (II) - 2016]

52. Which of the following equation is/are correct?

$$1. f(-2) = f(5)$$

$$2. f''(-2) + f''(0.5) + f'''(3) = 4$$

Select the correct answer using the codes given below:

- (a) Only 1 (b) Only 2
(c) Only 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

53. Let $f(x+y) = f(x)f(y)$ for all x and y . Then what is $f'(5)$ equal to [where $f'(x)$ is the derivative of $f(x)$]?]

- (a) $f(5)f'(0)$ (b) $f(5)-f'(0)$
(c) $f(5)f(0)$ (d) $f(5)+f'(0)$

[NDA (I) - 2017]

54. What is $\frac{d^2x}{dy^2}$ equal to?

- (a) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$ (b) $\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-2}$
(c) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$ (d) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

[NDA-2017(1)]

55. What is the derivative of $\log_{10}(5x^2+3)$ with respect to x ?

- (a) $\frac{x \log_{10} e}{5x^2+3}$ (b) $\frac{2x \log_{10} e}{5x^2+3}$
(c) $\frac{10x \log_{10} e}{5x^2+3}$ (d) $\frac{10x \log_e 10}{5x^2+3}$

[NDA-2017(1)]

56. If $y = (\cos x)^{(\cos x)^{(\cos x)^\infty}}$, then $\frac{dy}{dx}$ is equal to:

- (a) $\frac{-y^2 \tan x}{1 - y \ln(\cos x)}$ (b) $\frac{y^2 \tan x}{1 + y \ln(\cos x)}$
(c) $\frac{y^2 \tan x}{1 - y \ln(\sin x)}$ (d) $\frac{y^2 \sin x}{1 - y \ln(\sin x)}$

[NDA (II) - 2017]

57. A function is defined in $(0, \infty)$ by:

$$f(x) = \begin{cases} 1-x^2 & \text{for } 0 < x \leq 1 \\ \ln x & \text{for } 1 < x \leq 2 \\ \ln 2 - 1 + 0.5x & \text{for } 2 < x < \infty \end{cases}$$

Which one of the following is correct in respect of the derivative of the function, i.e., $f'(x)$?

- (a) $f'(x) = 2x$ for $0 < x \leq 1$
(b) $f'(x) = -2x$ for $0 < x \leq 1$
(c) $f'(x) = -2x$ for $0 < x < 1$
(d) $f'(x) = 0$ for $0 < x < \infty$

[NDA (II) - 2017]

58. If $y = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$, then $\frac{dy}{dx}$ is equal to:

- (a) 0 (b) 1
(c) $\frac{x-1}{x+1}$ (d) $\frac{x+1}{x-1}$

[NDA (II) - 2017]

59. Consider the following statement;

1. Derivative of $f(x)$ may not exist at some point.
 2. Derivative of $f(x)$ may exist finitely at some point.
 3. Derivative of $f(x)$ may be infinite (geometrically) at some point.
- Which of the following statements is / are correct?

- (a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2, and 3

[NDA (II) - 2017]

60. If $y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$, then $\frac{dy}{dx}$ is equal to:

- (a) $-\frac{2}{1+x^2}$ for all $|x| < 1$ (b) $-\frac{2}{1+x^2}$ for all $|x| > 1$
(c) $\frac{2}{1+x^2}$ for all $|x| < 1$ (d) None of these

[NDA (II) - 2017]

61. Consider the function $f(x) = \begin{cases} x^2 \ln|x| & x \neq 0 \\ 0 & x = 0 \end{cases}$ what is $f'(0)$ equal to?

- (a) 0 (b) 1
(c) -1 (d) It does not exist

[NDA (I) - 2018]

62. If $y = e^{x^2}$ then, what is $\frac{dy}{dx}$ at $x = \pi$ equal to

- (a) $(1 + \pi) e^{\pi^2}$ (b) $2\pi e^{\pi^2}$
(c) $2 e^{\pi^2}$ (d) e^{π^2}

[NDA (I) - 2018]

63. What is the derivative of the function $f(x) = e^{\tan x} + \ln(\sec x) - e^{\ln x}$ at $x = \frac{\pi}{4}$?

- (a) $e/2$ (b) e
(c) $2e$ (d) $4e$

[NDA (I) - 2018]

64. If $y = \tan^{-1}\left(\frac{5-2\tan\sqrt{x}}{2+5\tan\sqrt{x}}\right)$, then what is $\frac{dy}{dx}$ equal to?

- (a) $-\frac{1}{2\sqrt{x}}$ (b) 1
(c) -1 (d) $\frac{1}{2\sqrt{x}}$

[NDA (II) - 2018]

65. If $u = e^{ax} \sin bx$ and $v = e^{ax} \cos bx$, then what is $u \frac{du}{dx} + v \frac{dv}{dx}$ equal to?

- (a) $a e^{2ax}$ (b) $(a^2 + b^2) e^{ax}$
(c) $ab e^{2ax}$ (d) $(a + b) e^{ax}$

[NDA (II) - 2018]

66. If $y = |\sin x|^{|\sin x|}$, then what is the value of $\frac{dy}{dx}$ at $x = -\frac{\pi}{6}$?

- (a) $\frac{2^{-\frac{\pi}{6}}(6\ln 2 - \sqrt{3}\pi)}{6}$ (b) $\frac{2^{\frac{\pi}{6}}(6\ln 2 + \sqrt{3}\pi)}{6}$
(c) $\frac{2^{-\frac{\pi}{6}}(6\ln 2 + \sqrt{3}\pi)}{6}$ (d) $\frac{2^{\frac{\pi}{6}}(6\ln 2 - \sqrt{3}\pi)}{6}$

[NDA (II) - 2018]

67. What is $\frac{d\sqrt{1-\sin 2x}}{dx}$ equal to, where $\frac{\pi}{4} < x < \frac{\pi}{2}$?

- (a) $\cos x + \sin x$ (b) $-(\cos x + \sin x)$
(c) $\pm(\cos x + \sin x)$ (d) None of the above

[NDA (II) - 2018]

68. Let $f(x+y) = f(x)f(y)$ and where $f(x) = 1 + xg(x)\phi(x)$, $\lim_{x \rightarrow 0} g(x) = a$ and $\lim_{x \rightarrow 0} \phi(x) = b$. Then what is $f'(x)$ equal to?

- (a) $1 + abf(x)$ (b) $1 + ab$
(c) ab (d) $abf(x)$

69. Derivative of $\sec^2(\tan^{-1}x)$ w.r.t. x is
(a) $2x$ (b) $x^2 + 1$
(c) $x + 1$ (d) x^2 [NDA (II) - 2018]
70. If $f(x) = \sin(\cos x)$ then $f'(x) =$
(a) $\cos(\cos x)$ (b) $\sin(-\sin x)$
(c) $(\sin x)\cos(\cos x)$ (d) $(-\sin x)\cos(\cos x)$ [NDA (I) - 2019]
71. What is the derivative of $2^{(\sin x)^2}$ with respect to $\sin x$?
(a) $\sin x \cdot 2^{(\sin x)^2} \cdot \ln 4$ (b) $2 \sin x \cdot 2^{(\sin x)^2} \cdot \ln 4$
(c) $\ln(\sin x) 2^{(\sin x)^2}$ (d) $2 \sin x \cos x 2^{(\sin x)^2}$ [NDA (II) - 2019]
- Direction (for next two):**
Consider the equation $x^y = e^{x-y}$
72. What is $\frac{dy}{dx}$ at $x = 1$ equal to
(a) 0 (b) 1
(c) 2 (d) 4 [NDA (II) - 2019]
73. What is $\frac{d^2y}{dx^2}$ at $x = 1$ equal to?
(a) 0 (b) 1
(c) 2 (d) 4 [NDA (II) - 2019]
74. What is the derivative of $\tan^{-1}x$ with respect to $\cot^{-1}x$?
(a) -1 (b) 1
(c) $\frac{1}{1+x^2}$ (d) $\frac{x}{1+x^2}$ [NDA 2020]
75. If $x^m y^n = a^{m+n}$, then what is $\frac{dy}{dx}$ equal to?
(a) $\frac{my}{nx}$ (b) $-\frac{my}{nx}$
(c) $\frac{mx}{ny}$ (d) $-\frac{mx}{ny}$ [NDA 2020]
76. If $e^{\theta\phi} = c + 4\theta\phi$, where c is an arbitrary constant and ϕ is a function of θ , then what is $\phi d\theta$ equal to?
(a) $\theta d\phi$ (b) $-\theta d\phi$
(c) $4\theta d\phi$ (d) $-4\theta d\phi$ [NDA 2020]
77. What is the derivative of e^x w.r.t. x^e ?
(a) $\frac{xe^x}{e^x}$ (b) $\frac{e^x}{x^e}$
(c) $\frac{xe^x}{x^e}$ (d) $\frac{e^x}{e^x}$ [NDA (I) - 2021]
78. What is the derivative of $\sin(\ln x) + \cos(\ln x)$ with respect to x at $x = e$?
(a) $\frac{\cos 1 - \sin 1}{e}$ (b) $\frac{\sin 1 - \cos 1}{e}$
(c) $\frac{\cos 1 + \sin 1}{e}$ (d) 0 [NDA (I) - 2021]
79. If $x = e^t \cos t$ and $y = e^t \sin t$, then what is the value of dx/dy at $t = 0$ equal to?
(a) 0 (b) 1 (c) $2e$ (d) -1 [NDA (I) - 2021]
80. What is the derivative of e^{e^x} with respect to e^x ?
(a) e^{e^x} (b) ex
(c) $e^{e^x} ex$ (d) eex [NDA (II) - 2021]
81. If $y = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$ then what is $\frac{dy}{dx}$ at $x = 0$ equal to?
(a) 0 (b) 1
(c) 2 (d) 4 [NDA (II) - 2021]
82. If $y = \cos x \cdot \cos 4x \cdot \cos 8x$, then what is $\frac{1}{y} \frac{dy}{dx}$ at $x = \frac{\pi}{4}$ equal to?
(a) -1 (b) 0
(c) 1 (d) 3 [NDA (II) - 2021]
83. Let $f(x)$ be a polynomial function such that $f(f(x)) = x^4$. What is $f'(1)$?
(a) 0 (b) 1
(c) 2 (d) 4 [NDA (II) - 2021]
84. If $x^y y^x = 1$, then what is $\frac{dy}{dx}$ at $(1, 1)$ equal to?
(a) -1 (b) 0
(c) 1 (d) 4 [NDA (I) - 2022]
85. If $y = (x^x)^x$, then what is the value of $\frac{dy}{dx}$ at $x = 1$?
(a) $\frac{1}{2}$ (b) 1
(c) 2 (d) 4 [NDA (I) - 2022]
86. Let $y = [x + 1]$, $-4 < x < -3$ where $[.]$ is the greatest integer function. What is the derivative of y with respect to x at $x = -3.5$?
(a) -4 (b) -3.5
(c) -3 (d) 0 [NDA (I) - 2022]
87. If the derivative of the function $f(x) = \frac{m}{x} + 2nx + 1$ vanishes at $x = 2$, then what is the value of $m + 8n$?
(a) -2
(b) 0
(c) 2
(d) cannot be determined due to insufficient data [NDA (I) - 2022]
88. Let $f(x)$ be a function such that $f(x) = g(x)$ and $f''(x) = -f(x)$. Let $h(x) = \{f(x)\}^2 + \{g(x)\}^2$. Then consider the following statements:
1. $h'(3) = 0$ 2. $h(1) = h(2)$
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2 [NDA 2022 (II)]
89. If $y = \ln^2 \left(\frac{x^2 - x + 1}{x^2 + x + 1} \right)$, then what is $\frac{dy}{dx}$ at $x = 0$ equal to?
(a) -2 (b) 0
(c) 1 (d) 2 [NDA 2022 (II)]

90. If $\frac{d}{dx} \left(\frac{1+x^4+x^8}{1-x^2+x^4} \right) = ax + bx^3$, then which one of the following is correct?

(a) $a = b$ (b) $a = 2b$
(c) $a + b = 0$ (d) $2a = b$

[NDA 2022 (II)]

91. If $y = \frac{x\sqrt{x^2-16}}{2} - 8 \ln|x + \sqrt{x^2-16}|$, then what is $\frac{dy}{dx}$ equal to?

(a) $x\sqrt{x^2-16}$ (b) $x - \sqrt{x^2-16}$
(c) $\sqrt{x^2-16}$ (d) $4\sqrt{x^2-16}$

[NDA 2022 (II)]

92. What is the derivative of $\sin^2 x$ with respect of $\cos^2 x$?

(a) -1 (b) 1
(c) $\sin 2x$ (d) $\cos 2x$

[NDA - 2023 (1)]

93. What is the derivative of $\operatorname{cosec}(x^\circ)$?

(a) $-\operatorname{cosec}(x^\circ) \cot(x^\circ)$ (b) $-\frac{\pi}{180} \operatorname{cosec}(x^\circ) \cot(x^\circ)$
(c) $\frac{\pi}{180} \operatorname{cosec}(x^\circ) \cot(x^\circ)$ (d) $-\frac{\pi}{180} \operatorname{cosec}(x) \cot(x)$

[NDA - 2023 (1)]

Consider the following for the next three (03) items that follow:

Let $f(x) = |\ln x|$, $x \neq 1$.

94. What is the derivative of $f(x)$ at $x = 0.5$?

(a) -2 (b) -1
(c) 1 (d) 2

[NDA - 2023 (1)]

95. What is the derivative of $f(x)$ at $x = 2$?

(a) $-1/2$ (b) -1
(c) $1/2$ (d) 2

[NDA - 2023 (1)]

96. What is the derivative of $\operatorname{fof}(x)$, where $1 < x < 2$?

(a) $\frac{1}{\ln x}$ (b) $\frac{1}{x \ln x}$
(c) $-\frac{1}{\ln x}$ (d) $-\frac{1}{x \ln x}$

[NDA - 2023 (1)]

97. If $f(x) = |\ln|x||$ where $0 < x < 1$, then what is $f'(0.5)$ equal to?

(a) -2 (b) -1
(c) 0 (d) 2

[NDA-2023 (2)]

98. If $f'(x) = \cos(\ln x)$ and $y = f\left(\frac{2x-3}{x}\right)$, then what is $\frac{dy}{dx}$ equal to?

(a) $\cos\left(\ln\left(\frac{2x-3}{x}\right)\right)$ (b) $-3/x^2 \sin\left(\ln\left(\frac{2x-3}{x}\right)\right)$
(c) $3/x^2 \cos\left(\ln\left(\frac{2x-3}{x}\right)\right)$ (d) $-3/x^2 \cos\left(\ln\left(\frac{2x-3}{x}\right)\right)$

[NDA-2023 (2)]

Consider the following for the next two (02) items that follow:

Let $f(x)$ and $g(x)$ be two functions such that $g(x) = x - \frac{1}{x}$

and $\operatorname{fog}(x) = x^3 - \frac{1}{x^3}$.

99. What is $g[f(x) - 3x]$ equal to?

(a) $x^3 - \frac{1}{x^3}$ (b) $x^3 + \frac{1}{x^3}$
(c) $x^2 - \frac{1}{x^2}$ (d) $x^2 + \frac{1}{x^2}$

[NDA-2024 (1)]

100. What is $f''(x)$ equal to?

(a) $-\frac{2}{x^3}$ (b) $2x + \frac{2}{x^3}$
(c) $6x + 3$ (d) $6x$

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $h(x) = f(2f(x)+2)$ and $g(x) = (h(x))^2$

101. What is $h'(0)$ equal to?

(a) -2 (b) -1
(c) 0 (d) 2

[NDA - 2024 (2)]

102. What is $g'(0)$ equal to?

(a) -4 (b) -2
(c) 0 (d) 4

[NDA - 2024 (2)]

Consider the following for the two (02) items that follow:

Let $x = \sec\theta - \cos\theta$ and $y = \sec^4\theta - \cos^4\theta$.

103. What is $\left(\frac{dy}{dx}\right)^2$ equal to?

(a) $\frac{4(y^2+4)}{(x^2+4)}$ (b) $\frac{4(y^2-4)}{(x^2-4)}$
(c) $\frac{16(y^2+4)}{(x^2+4)}$ (d) $\frac{16(y^2-4)}{(x^2-4)}$

[NDA - 2025 (1)]

104. What is $\left(\frac{x^2+4}{y^2+4}\right) \frac{dy}{dx} \left[(x^2+4) \frac{d^2y}{dx^2} - 16y \right]$ equal to ?

(a) $16x$ (b) $16y$
(c) $-16x$ (d) $-16y$

[NDA - 2025 (1)]

Consider the following for the two (02) items that follow:

Let $(x+y)^{p+q} = x^p \cdot y^q$, where p, q are positive integers.

105. The derivative of y with respect to x

(a) depends on p only
(b) depends on q only
(c) depends on both p and q
(d) is independent of both p and q

[NDA - 2025 (1)]

106. If $p + q = 10$, then what is $\left(\frac{dy}{dx}\right)$ equal to?

(a) $\frac{y}{x}$ (b) xy
(c) $x^{10}y^{10}$ (d) $\left(\frac{y}{x}\right)^{10}$

[NDA - 2025 (1)]

Consider the following for the two (02) items that follow:

Let $y = \sin^{-1}\left(x - \frac{4x^3}{27}\right)$

107. What is y equal to?

(a) $y = \sin^{-1} x$

(b) $y = \sin^{-1}\left(\frac{x}{3}\right)$

(c) $y = 3\sin^{-1} x$

(d) $y = 3\sin^{-1}\left(\frac{x}{3}\right)$

[NDA – 2025 (1)]

108. What is $\frac{dy}{dx}$ equal to?

(a) $\frac{1}{\sqrt{9-x^2}}$

(b) $\frac{1}{\sqrt{3-x^2}}$

(c) $\frac{3}{\sqrt{9-x^2}}$

(d) $\frac{9}{\sqrt{9-x^2}}$

[NDA – 2025 (1)]

For the following **three** items:

Let $y = f(x) = \frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \ln \sqrt{1-x^2}$

109. What is the slope of the tangent to the curve $y = f(x)$ at $x = 0.5$?

(a) $4\pi\sqrt{3}/27$

(b) $8\pi\sqrt{3}/27$

(c) 4π

(d) 8π

[NDA-2025 (2)]

110. What is $\frac{d^2y}{dx^2}$ at $x = 0$ equal to ?

(a) 0

(b) 0.5

(c) 1

(d) 1.5

[NDA-2025 (2)]

111. If $x = \sin\theta$, then what is $\frac{dy}{dx}$ equal to?

(a) $\theta \sec\theta$

(b) $\theta \sec^2\theta$

(c) $\theta \sec^3\theta$

(d) $2\tan\theta + \theta \sec^2\theta$

[NDA-2025 (2)]

For the following next **one** items:

Consider the function

$$f(x) = \begin{cases} 4(5^x) & x < 0 \\ 8k + x & x \geq 0 \end{cases}$$

112. What is $f'(-1)$ equal to?

(a) $\frac{2}{5} \ln 5$

(b) $\frac{3}{5} \ln 5$

(c) $\frac{4}{5} \ln 5$

(d) $20 \ln 5$

[NDA-2025 (2)]

For the following **one** items:

Let the function $f(x) = |x - 3| + |x - 4|$ be defined on the interval $[0, 5]$.

113. What is $\frac{dy}{dx}$ at $x = 3.5$ equal to?

(a) 0

(b) 1

(c) 2

(d) 3.5

[NDA-2025 (2)]

ANSWER KEY

1.	a	2.	b	3.	d	4.	d	5.	b	6.	b	7.	c	8.	d	9.	c	10.	b
11.	c	12.	c	13.	a	14.	a	15.	b	16.	a	17.	b	18.	c	19.	c	20.	b
21.	b	22.	c	23.	a	24.	c	25.	c	26.	b	27.	a	28.	c	29.	c	30.	a
31.	b	32.	b	33.	d	34.	d	35.	a	36.	c	37.	a	38.	d	39.	c	40.	b
41.	d	42.	d	43.	d	44.	a	45.	d	46.	a	47.	c	48.	c	49.	d	50.	a
51.	d	52.	a	53.	a	54.	c	55.	c	56.	a	57.	c	58.	a	59.	d	60.	a
61.	a	62.	b	63.	c	64.	a	65.	a	66.	a	67.	a	68.	d	69.	a	70.	d
71.	a	72.	a	73.	b	74.	a	75.	b	76.	b	77.	a	78.	a	79.	b	80.	a
81.	b	82.	a	83.	c	84.	a	85.	b	86.	d	87.	d	88.	c	89.	b	90.	d
91.	c	92.	a	93.	b	94.	a	95.	c	96.	d	97.	a	98.	c	99.	a	100.	d
101.	d	102.	a	103.	c	104.	c	105.	d	106.	a	107.	d	108.	c	109.	a	110.	c
111.	c	112.	c	113.	a														

Solutions

Sol.1. (a)

$$f(x) = 2^x$$

$$f'(x) = 2^x \log_e 2$$

$$f''(x) = 2^x \log_e 2 \cdot 2 \log_e 2 = 2^x (\log_e 2)^2$$

Sol.2. (b)

$$y = (1 + x^{1/4})(1 + x^{1/2})(1 - x^{1/4})$$

$$= (1 + x^{1/2})(1 - x^{1/2}) = 1 - x$$

$$\therefore \frac{dy}{dx} = -1$$

Sol.3. (d)

$$y = \ln \sqrt{\tan x}$$

$$y = \frac{1}{\sqrt{\tan x}} \cdot \frac{1}{2\sqrt{\tan x}} \sec^2 x$$

put $x = \pi/4$

$$y = 1$$

Sol.4. (d)

Given that, $y = \cot t$ and $x = \sin t$

Then, $\frac{dy}{dt} = -\sin t$, and $\frac{dx}{dt} = \cos t$

Now, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-\sin t}{\cos t} = \frac{-x}{y}$

Sol.5. (b)

We have, $y = \frac{x+1}{x-1}$

Now, differentiating w. r. t. x., we get

$$\frac{dy}{dx} = \frac{(x-1) \frac{d}{dx}(x+1) - (x+1) \frac{d}{dx}(x-1)}{(x-1)^2}$$

$$= \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

Sol.6. (b)

Let $u = \sqrt{x^2 + 16}$, and $v = x^2$

Now, $\frac{du}{dx} = \frac{1}{2\sqrt{x^2 + 16}} \times 2x$

$$= \frac{x}{\sqrt{x^2 + 16}}$$

and $\frac{dv}{dx} = 2x$

Now, rate of change of u w. r. t. x., is

$$\frac{dx}{dv} = \frac{du/dx}{dv/dx} = \frac{x}{\sqrt{x^2 + 16}} \times \frac{1}{2x}$$

$$\frac{du}{dv} = \frac{1}{2\sqrt{x^2 + 16}}$$

$$\frac{du}{dv} \text{ at } (x=3) = \frac{1}{2\sqrt{9+16}} = \frac{1}{2\sqrt{25}} = \frac{1}{2 \times 5} = \frac{1}{10}$$

$$\frac{d\sqrt{x^2+16}}{d(x^2)} = \frac{1}{10} \text{ at } x = 3$$

Sol.7. (c)

$$f(x) = x^2 - 6x + 8$$

$$f'(x) = 2x - 6$$

$$f'(c) = 2c - 6 = 0$$

$c = 3$ that is lie between $[2, 4]$

Sol.8. (d)

$$f(x) = 2^{\sin x}$$

$$f'(x) = 2^{\sin x} (\ln 2) \cos x$$

Sol.9. (c)

Given, $x^m + y^m = 1$

On differentiating both sides w. r. t. x., we get

$$mx^{m-1} + my^{m-1} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{mx^{m-1}}{my^{m-1}}$$

$$= -\frac{x^{m-1}}{y^{m-1}} = -\left(\frac{x^m}{y^m}\right)\left(\frac{y}{x}\right)$$

Given, $\frac{-x}{y} = -\left(\frac{x^m}{y^m}\right)\left(\frac{y}{x}\right)$

$$\therefore \left(\frac{x}{y}\right)^{m-2} = 1$$

Which is true when $m = 2$.

Sol.10. (b)

$y = \ln(e^{mx} + e^{-mx})$

On differentiating it w. r. t. x., we get

$$\frac{dy}{dx} = \frac{1}{e^{mx} + e^{-mx}} \cdot \frac{d}{dx}(e^{mx} + e^{-mx})$$

$$= \frac{1}{e^{mx} + e^{-mx}} (me^{mx} - me^{-mx}) = \left[\frac{m(e^{mx} - e^{-mx})}{e^{mx} + e^{-mx}} \right]$$

$$\therefore \left(\frac{dy}{dx}\right)_{\text{at } x=0} = m \left(\frac{1-1}{1+1}\right) = 0$$

Sol.11. (c)

Statement I.

Given, $y = \ln(\sec x + \tan x)$

On differentiating it w. r. t. x., we get

$$\frac{dy}{dx} = \frac{1}{(\sec x + \tan x)} (\sec x + \tan x)$$

$$= \frac{1}{(\sec x + \tan x)} (\sec x \cdot \tan x + \sec^2 x)$$

$$= \frac{1}{(\sec x + \tan x)} \sec x (\tan x + \sec x) = \sec x$$

Statement II.

Given, $y = \log(\operatorname{cosec} x - \cot x)$

$$\frac{dy}{dx} = \frac{1}{(\operatorname{cosec} x - \cot x)} \frac{d}{dx}(\operatorname{cosec} x - \cot x)$$

$$= \frac{1}{(\operatorname{cosec} x - \cot x)} (-\operatorname{cosec} x \cdot \cot x + \operatorname{cosec}^2 x)$$

$$= \operatorname{cosec} x \cdot \frac{(\operatorname{cosec} x - \cot x)}{(\operatorname{cosec} x)} = \operatorname{cosec} x$$

So, statements I and II both are true.

Sol.12. (c)

Given curve is $2x^3 - 3y^2 = 7$, $y \neq 0$.

Now, differentiating both sides w. r. t. x, we get

$$2 \cdot 3 \cdot x^2 - 3 \cdot 2 \cdot y \frac{dy}{dx} = 0$$

$$\Rightarrow x^2 - y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x^2}{y}$$

Sol.13. (a)

Let $u = \sec^2 x$ and $v = \tan^2 x$

Now, $\frac{du}{dx} = 2 \sec x \cdot \sec x \cdot \tan x = 2 \sec^2 x \cdot \tan x$

and $\frac{dv}{dx} = 2 \tan x \cdot \sec^2 x$

$$\therefore \frac{du}{dv} = \frac{d(\sec^2 x)}{d(\tan^2 x)} = \frac{du/dx}{dv/dx} = \frac{2 \sec^2 x \cdot \tan x}{2 \tan x \cdot \sec^2 x} = 1$$

Sol.14. (a)

Let $u = \log_x x = 1$

$$\Rightarrow \frac{du}{dx} = 0, \text{ and } v = \log x$$

$$\Rightarrow \frac{dv}{dx} = \frac{1}{x}$$

$$\therefore \frac{du}{dv} = \frac{du/dx}{dv/dx} = 0$$

Sol.15. (b)

Given curve, $y = x^x$

Taking log on both sides,

$$\log y = x \log x$$

on differentiating both sides w. r. t. x., we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x$$

$$\Rightarrow \frac{dy}{dx} = (1 + \log x) \cdot y$$

$$\Rightarrow \frac{dy}{dx} = x^x (1 + \log x)$$

$$\therefore \left(\frac{dy}{dx}\right)_{\text{at } x=1} = (1)^1 (1 + \log 1) = 1(1+0) = 1$$

Sol.16. (a)

Given curve,

$$y = \sin(ax + b), \text{ where } a \text{ and } b \text{ are constants, and } a \neq 0.$$

Now, differentiating both sides w. r. t. x., we get

$$\frac{dy}{dx} = \cos(ax + b) \cdot a$$

Again, differentiating both sides w. r. t. x., we get

$$\frac{d^2 y}{dx^2} = -\sin(ax + b) \cdot a \cdot a = -a^2 \sin(ax + b)$$

$$= -a^2 y.$$

$$[\because y = \sin(ax + b)]$$

$$\therefore \left(\frac{d^2 y}{dx^2}\right)_{\text{at } x = -\frac{b}{a}} = -a^2 \sin\left\{a \cdot \left(-\frac{b}{a}\right) + b\right\}$$

$$= -a^2 \cdot \sin(-b + b) = -a^2 \cdot \sin 0 = -a^2 \cdot 0 = 0$$

Sol.17. (b)

Let $f(x) = x^3$ and $g(x) = x^2$

Now, $\frac{df(x)}{dx} = 3x^2$ and $\frac{dg(x)}{dx} = 2x$

$$\therefore \text{Derivative of } x^3 \text{ w. r. t. } x^2$$

$$\frac{df(x)}{dg(x)} = \left\{ \frac{df(x)}{dx} \right\} \times \left\{ \frac{dx}{dg(x)} \right\} = 3x^2 \times \frac{1}{2x} = \frac{3x}{2}$$

Sol.18. (c)

Let $f(x) = \sin(\sin x)$

On differentiating w. r. t. t. x, we get

$$f'(x) = \cos(\sin x) \cdot \frac{d}{dx}(\sin x)$$

$$= \cos x \cdot \cos(\sin x)$$

Sol.19. (c)

Let $f(x) = |x - 1|$

Redefined the function $f(x)$,

$$f(x) = \begin{cases} 1-x, & x < 1 \\ x-1, & x > 1 \end{cases} \Rightarrow f'(x) = \begin{cases} -1, & x < 1 \\ 1, & x > 1 \end{cases}$$

$$\therefore f'(2) = 1$$

Sol.20. (b)

$$f(x) = 2x^2 + 3x - 5,$$

$$f'(x) = 4x + 3$$

$$f'(0) + 3f'(-1) = 3 + 3(-1) = 0$$

Sol.21. (b)

$$y = \sqrt{\frac{1+\cos x}{1-\cos x}} = \sqrt{\frac{2\cos^2 \frac{x}{2}}{2\sin^2 \frac{x}{2}}} = \cot \frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{1}{2} \operatorname{cosec}^2 \frac{x}{2}$$

Sol.22. (c)

$$f(x) = x^2$$

$$z = f \circ f(x) = x^4$$

$$z'(x) = 4x^3$$

Sol.23. (a)

$$\text{We have, } x = a(\cos \theta + \theta \sin \theta)$$

$$\text{and } y = a(\sin \theta - \theta \cos \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = a(-\sin \theta + \theta \cos \theta + \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = a \theta \cos \theta$$

$$\text{and } \frac{dy}{d\theta} = a(\cos \theta + \theta \sin \theta - \cos \theta)$$

$$\Rightarrow \frac{dy}{d\theta} = a \theta \sin \theta$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a \theta \sin \theta}{a \theta \cos \theta} = \tan \theta$$

Sol.24. (c)

$$\text{We have, } \frac{dy}{dx} = \tan \theta$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \sec^2 \theta \frac{d\theta}{dx}$$

$$\Rightarrow \frac{d^2 y}{dx^2} \sec^2 \theta \left(\frac{1}{a \theta \cos \theta} \right) \left(\because \frac{dx}{d\theta} = a \theta \cos \theta \right)$$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{\sec^3 \theta}{a \theta}$$

Sol.25. (c)

$$\text{Given, } y = x \log x + x e^x$$

On differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + \log x + x e^x + e^x$$

$$\Rightarrow \frac{dy}{dx} = 1 + \log x + x e^x + e^x$$

$$\therefore \left(\frac{dy}{dx} \right)_{x=1} = 1 + \log 1 + 1 \cdot e^1 + 1$$

$$= 1 + 2e \quad (\because \log 1 = 0)$$

Sol.26. (b)

$$f(x) = ax^2 + bx + c$$

$$f(1) = f(-1)$$

$$a + b + c = a - b + c$$

$$2b = 0$$

$$b = 0$$

Sol.27. (a)

$$f(x) = ax^2 + bx + c$$

$$f'(x) = 2ax + b$$

$$f'(a) = 2aa + b, f'(b) = 2ab + b, f'(c) = 2ac + b$$

given that a, b, c are in AP

than after multiply with 2a

2aa, 2ab, 2ac will be in AP

now add b in each term

2aa + b, 2ab + b, 2ac + b will be in AP

Sol.28. (c)

$$f''(x) = 2a$$

$$f''(a) = f''(b) = f''(c) = 2a$$

so answer is AP and GP both

Solutions (for next two)

$$\text{We have, } \frac{d}{dx} \left(\frac{1+x^2+x^4}{1+x+x^2} \right) = Ax + B \dots (i)$$

Let us first divide x^4+x^2+1 by x^2+x+1

$$\text{Thus, } \frac{x^4+x^2+1}{x^2+x+1} = x^2 - x + 1$$

$\therefore$ From equation (i), we have

$$\frac{d}{dx} (x^2 - x + 1) = Ax + B$$

$$\Rightarrow 2x - 1 = Ax + B \dots (ii)$$

Sol.29. (c)

On comparing, the coefficient of A, we get $A = 2$

Sol.30. (a)

On comparing the coefficient of constant terms in equation (ii), we get $B = -1$.

Sol.31. (b)

$$\text{Let } u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$\text{and } v = \tan^{-1} x$$

$$\text{To find : } \frac{du}{dv}$$

$$\text{Consider, } u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

On putting $x = \tan \theta$, we get

$$u = \tan^{-1} \left(\frac{\sqrt{1+\tan^2 \theta}-1}{\tan \theta} \right)$$

$$= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right)$$

$$= \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2}$$

$$\Rightarrow u = \frac{1}{2} \tan^{-1} x$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{2} \left(\frac{1}{1+x^2} \right) \dots (i)$$

Now consider $v = \tan^{-1} x$

$$\Rightarrow \frac{dv}{dx} = \frac{1}{1+x^2} \dots (ii)$$

$$\text{We have, } \frac{du}{dv} = \frac{du}{dx} \times \frac{dx}{dv} \text{ [by using equation (i)]}$$

and (ii)

$$= \frac{1}{2} \left(\frac{1}{1+x^2} \right) \cdot (1+x^2) \cdot \frac{1}{2}$$

Sol.32. (b)

$$x = \frac{a(1-t^2)}{1+t^2} \quad y = \frac{2at}{1+t^2}$$

after squaring and addition of both equations

$$x^2 + y^2 = a^2$$

this equation represent a circle.

Sol.33. (d)

$$x^2 + y^2 = a^2$$

diff. both side w.r.t. x

$$2x + 2yy' = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Sol.34. (d)

$$\frac{d^2 y}{dx^2} = -\frac{y - xy'}{y^2} = -\frac{y - x(-x/y)}{y^2}$$

$$\frac{d^2 y}{dx^2} = -\frac{a^2}{y^2}$$

Sol.35. (a)

$$y = \cot^{-1} \left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right]$$

$$\text{or } y = \cot^{-1}$$

$$\left[\frac{(\sqrt{1+\sin x} + \sqrt{1-\sin x})^2}{(\sqrt{1+\sin x} - \sqrt{1-\sin x})(\sqrt{1+\sin x} + \sqrt{1-\sin x})} \right]$$

$$\text{or } y = \cot^{-1} \left[\frac{1 + \sin x + 1 - \sin x + 2\sqrt{1-\sin^2 x}}{1 + \sin x - 1 + \sin x} \right]$$

$$\text{or } y = \cot^{-1} \left[\frac{2(1 + \cos x)}{2 \sin x} \right]$$

$$= \cot^{-1} \left[\frac{2 \cos^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \right]$$

$$\text{or } y = \cot^{-1} \left[\cot \frac{x}{2} \right]$$

$$\text{on differentiating } y = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

Sol.36. (c)

$$s = \sqrt{t^2 + 1} \Rightarrow s^2 = t^2 + 1$$

$$2s \frac{ds}{dt} = 2t$$

$$\frac{ds}{dt} = \frac{t}{s} = \frac{t}{\sqrt{t^2 + 1}}$$

$$\frac{d^2 s}{dt^2} = \frac{\sqrt{1+t^2} - \frac{t \times 2t}{2\sqrt{t^2+1}}}{(\sqrt{t^2+1})^2}$$

$$= \frac{t^2 + 1 - t^2}{\sqrt{t^2+1} \cdot (t^2+1)} = \frac{1}{s \cdot s^2} = \frac{1}{s^3}$$

Sol.37. (a)

$$y_1 = \log(x + \sin x)$$

$$y_2 = (x + \cos x)$$

$$\frac{dy_1}{dx} = \frac{1}{(x + \sin x)} (1 + \cos x) = \frac{(1 + \cos x)}{(x + \sin x)} \dots (i)$$

$$\frac{dy_2}{dx} = (1 - \sin x) \dots (ii)$$

Dividing equation (i) by (ii), we get

$$= \frac{\frac{dy_1}{dx}}{\frac{dy_2}{dx}} = \frac{1 + \cos x}{(1 - \sin x)} = \frac{(1 + \cos x)}{(x + \sin x)(1 - \sin x)}$$

Sol.38. (d)

$$x^a y^b = (x - y)^{a+b}$$

taking log both side

$$a \log x + b \log y = (a + b) \log(x - y)$$

differentiate w.r.t. x

$$\frac{a}{x} + \frac{b}{y} \frac{dy}{dx} = (a + b) \cdot \frac{1}{x - y} \left(1 - \frac{dy}{dx} \right)$$

by solving it

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\frac{dy}{dx} - \frac{y}{x} = 0$$

Sol.39. (c)

$$f(x) = |x^2 - 5x + 6|$$

$$f'(x) = |2x - 5|$$

$$f'(4) = |2 \cdot 4 - 5|$$

$$f'(4) = |8 - 5|$$

$$f'(4) = |3| = 3$$

Sol.40. (b)

$$f''(x) = |2|$$

$$f''(2.5) = 2$$

Sol.41. (d)

This function is continuous for all values of x and not differentiable at origin.

Sol.42. (d)

For $x < 0$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{3}(-h) - 0}{-h} = \lim_{h \rightarrow 0} \frac{1}{3}$$

$$\frac{dy}{dx} = \frac{1}{3}$$

Sol.43. (d)

$$f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

Expanding corresponding to C_3

$$f(x) = \cos x (6p^2 + p) + 0 + p^3$$

$$(-x^3 - 6\sin x)$$

On differentiating w. r. t. to x .

$$f'(x) = -\sin x (6p^2 + p) - p^3 (3x^2 + 6\cos x)$$

...(i)

Put $x = 0$

$$f'(0) = 0 - p^3 = (0 + 6 \cos 0^\circ)$$

$$f'(0) = -p^3 (6)$$

$$f'(0) = -6p^3$$

Sol.44. (a)

Again differentiation of equation (i)

$$f''(x) = -\cos x (6p^2 + p) - p^3 (6x - 6\sin x)$$

$$f''(x) = -\cos x (6p^2 + p) - 6p^3 (x - \sin x)$$

Putting $x = 0$

$$f''(0) = -\cos 0^\circ (6p^2 + p) - 6p^3 (0 - \sin 0)$$

$$f''(0) = -(6p^2 + p) - 6p^3 (0)$$

$$f''(0) = -(6p^2 + p)$$

Since, $f''(0) = 0$

$$-(6p^2 + p) = 0$$

$$p(6p + 1) = 0$$

$$p = 0, p = -\frac{1}{6}$$

Solutions (for next three)

$$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$$

On differentiating w. r. t. 'x'

$$f'(x) = 3x^2 + 2x f'(1) + f''(2) \quad \dots(i)$$

$$f''(x) = 6x + 2f'(1) \quad \dots(ii)$$

$$f'''(x) = 6 \quad \dots(iii)$$

Putting $x = 1$ in equation (ii) and $x = 3$ in equation (iii)

$$f'(1) = 3 + 2 f'(1) + f''(2)$$

$$\Rightarrow f'(1) + f''(2) = -3 \quad \dots(iv)$$

$$\text{Now, } f''(2) = 12 + 2f'(1) \quad \dots(v)$$

From equations (iv) and (v)

$$f'(1) + 12 + 2 f'(1) = -3$$

$$3f'(1) = -15$$

$$f'(1) = -5$$

Sol.45. (d)

Putting this value in equation (v)

$$f''(2) = 12 + 2(-5)$$

$$f''(2) = 2$$

$$\text{and } f'''(3) = 6$$

Now the $f(x)$ is

$$f(x) = x^3 + x^2(-5) + 2x + 6$$

$$f(x) = x^3 - 5x^2 + 2x + 6$$

Sol.46. (a)

Put $x = 1$ in $f(x)$

$$f(1) = 1 - 5 + 2 + 6$$

$$f(1) = 4$$

Put $x = 2$

$$f(2) = 8 - 20 + 4 + 6$$

$$f(2) = -2$$

Sol.47. (c)

From equation (iii)

$$f'''(10) = 6$$

$$\text{Now, } f(0) = 6$$

Sol.48. (c)

$$\text{I. } f(2) = -2$$

$$f(1) - f(0) = 4 - 6 = -2$$

$$\Rightarrow f(2) = f(1) - f(0) \quad (\text{True})$$

$$\text{II. } f'(2) - 2 f'(1) = 2 - 2(-5)$$

$$= 2 + 10$$

$$= 12$$

Sol.49. (d)

$$y = \log_{10} x + \log_x 10 + \log_x x + \log_{10} 10$$

$$y = \log_{10} x + \frac{1}{\log_{10} x} + 1 + 1$$

$$y = \log_{10} x + (\log_{10} x)^{-1} + 2$$

$$\frac{dy}{dx} = \frac{1}{x} \log_{10} e - (\log_{10} x)^{-2} \cdot \frac{1}{x} \log_{10} e + 0$$

$$\frac{dy}{dx} = \frac{1}{x} \log_{10} e - (\log_{10} x)^{-2} \cdot \frac{1}{x} \log_{10} e$$

At $x = 10$,

$$\left(\frac{dy}{dx}\right)_{x=10} = \frac{1}{10} \log_{10} 10 - (\log_{10} 10)^{-2} \cdot \frac{1}{10} \log_{10} e$$

$$= \frac{1}{10} \cdot 1 - (1)^{-2} \cdot \frac{1}{10} = \frac{1}{10} - \frac{1}{10} = 0$$

Sol.50. (a)

$$\text{We have, } f(x) = (|x| - (|x-1|)^2)$$

$$\text{When } x > 1, f(x) = [x - (x-1)]^2 = 1$$

$$\therefore f'(x) = 0$$

Sol.51. (d)

$$\therefore f(x) = (|x| - |x-1|)^2$$

When $0 < x < 1$

$$\therefore f(x) = [x - (x-1)]^2 = (2x-1)^2$$

$$\therefore f'(x) = 2(2x-1)(2) = 8x-4$$

Sol.52. (a)

$$\therefore |x| - |x-1| = \begin{cases} -1, & x < 0 \\ 2x-1 & 0 \leq x < 1 \\ -1, & 1 \leq x \end{cases}$$

$$\therefore f(x) = \begin{cases} 1, & \text{when } 0 > x \text{ or } x \geq 1 \\ (2x-1)^2, & \text{when } 0 \leq x < 1 \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} 0, & \text{when } 0 > x \text{ or } x \geq 1 \\ 4(2x-1), & \text{when } 0 \leq x < 1 \end{cases}$$

$$\Rightarrow f''(x) = \begin{cases} 0, & \text{when } 0 > x \text{ or } x \geq 1 \\ 8, & \text{when } 0 \leq x < 1 \end{cases}$$

$$\text{I. } f(-2) = 1 \text{ and } f(5) = 1$$

$$\text{So, } f(-2) = f(5) = 1$$

$\Rightarrow$ Statement I is correct

$$\text{II. } f''(-20) = 0$$

$$f''(0.5) = 8 \text{ and } f''(3) = 0$$

$$\text{Then, } f''(-2) + f''(0.5) + f''(3) = 0 + 8 + 0 = 8$$

So, statement II incorrect.

Sol.53. (a)

$$f(x+y) = f(x) \cdot f(y)$$

$$f(0) = f(0+0) = f(0) \cdot f(0)$$

$$\text{or } f'(0) = f(0) \cdot f'(0) + f(0) \cdot f'(0) \{\text{by product rule}\}$$

$$\text{or } f'(0) = 2f(0) \cdot f'(0)$$

$$\text{or } f(0) = 0$$

$$\text{Now } f(5) = f(0+5) = f(0) \cdot f(5)$$

$$\Rightarrow f(5) = f(0) \cdot f(5) + f(5) \cdot f'(0)$$

$$\Rightarrow f(5) = 0 + f(5) \cdot f'(0) \quad \{\text{from (i)}\}$$

$$\Rightarrow f'(5) = f(5) \cdot f'(0)$$

Sol.54. (c)

$$\text{let } y = f(x), p = \frac{dy}{dx} \text{ and } q = \frac{d^2y}{dx^2},$$

$$\frac{dx}{dy} = \frac{1}{p} \Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{p^2} \frac{dp}{dy}$$

$$\Rightarrow \frac{d^2x}{dy^2} = -\frac{1}{p^2} \frac{dp}{dy} = -\frac{1}{p^2} \frac{dp}{dx} \frac{dx}{dy} = -\frac{q}{p^3}$$

$$= -\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$$

Sol.55. (c)

$$f(x) = \frac{-2x(-e^{-x^2})}{2\sqrt{1-e^{-x^2}}} = \frac{xe^{-x^2}}{\sqrt{1-e^{-x^2}}}$$

which is defined $\forall x \in \mathbb{R}$, except $x = 0$

$$\Rightarrow f(x) \text{ is differentiable on } (-\infty, 0) \cup (0, \infty)$$

Sol.56. (a)

$$y = (\cos x)^{(\cos x)^{(\cos x)^{\dots \infty}}}$$

$$\dots(i)$$

$$\text{or } y = (\cos x)^y$$

Taking log on both the sides,

$$\log y = y \cdot \log(\cos x)$$

differentiating w. r. t. x , we get

$$\frac{1}{y} \frac{dy}{dx} = y \left(\frac{-\sin x}{\cos x} \right) + \log(\cos x) \cdot \frac{dy}{dx}$$

$$\text{or } \frac{dy}{dx} \left[\log(\cos x) - \frac{1}{y} \right] = y \tan x$$

$$\text{or } \frac{dy}{dx} = \frac{y^2 \tan x}{y \cdot \log(\cos x) - 1} = \frac{-y^2 \tan x}{1 - y \cdot \log(\cos x)}$$

Sol.57. (c)

$$f(x) = \begin{cases} 1-x^2 & \text{for } 0 < x \leq 1 \\ \ln x & \text{for } 1 < x \leq 2 \\ \ln 2 - 1 + 0.5x & \text{for } 2 < x < 0 \end{cases}$$

$$f'(x) = \begin{cases} -2x & \text{for } 0 < x \leq 1 \\ 1/x & \text{for } 1 < x \leq 2 \\ 0.5 & \text{for } 2 < x < 0 \end{cases}$$

function is not differentiable at $x = 1$

$$\text{so } f(x) = -2x \text{ for } 0 < x < 1$$

Sol.58. (a)

$$y = \sec^{-1} \left(\frac{x+1}{x-1} \right) + \sin^{-1} \left(\frac{x-1}{x+1} \right)$$

$$= \cos^{-1} \left(\frac{x-1}{x+1} \right) + \sin^{-1} \left(\frac{x-1}{x+1} \right)$$

$$= \frac{\pi}{2} \left\{ \because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right\}$$

$$\therefore \frac{dy}{dx} = 0$$

Sol.59. (d)

All statements are correct.

Sol.60. (a)

$$y = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\text{Put } x = \tan \theta,$$

$$y = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$\text{Put } x = \tan \theta,$$

$$y = \cos^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) = \cos^{-1} (\sin 2\theta)$$

$$= \cos^{-1} \cos \left(\frac{\pi}{2} - 2\theta \right)$$

$$= \frac{\pi}{2} - 2\theta \left\{ \text{where } 0 \leq \left(\frac{\pi}{2} - 2\theta \right) \leq \pi \right.$$

Sol.61. (a)

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \log h}{h} = 0$$

Sol.62. (b)

$$y = e^{x^2}$$

$$y' = 2xe^{x^2}$$

$$\text{at } x = \pi, y' = 2\pi e^{\pi^2}$$

Sol.63. (c)

$$f(x) = e^{\tan x} + \ln(\sec x) - x$$

$$f'(x) = e^{\tan x} \cdot \sec^2 x + \frac{\sec x \tan x}{\sec x} - 1$$

$$f'(\pi/4) = 2e + 1 - 1 = 2e$$

Sol.64. (a)

$$y = \tan^{-1} \left(\frac{5 - 2 \tan \sqrt{x}}{2 + 5 \tan \sqrt{x}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{5}{2} - \tan \sqrt{x}}{1 + \left(\frac{5}{2} \right) \tan \sqrt{x}} \right)$$

$$= \tan^{-1} \frac{5}{2} - \tan^{-1} \tan \sqrt{x} = \tan^{-1} \frac{5}{2} - \sqrt{x}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$$

Sol.65. (a)

$$u \frac{du}{dx} + v \frac{dv}{dx}$$

$$= e^{ax} \sin bx [ae^{ax} \sin bx + be^{ax} \cos bx]$$

$$+ e^{ax} \cos bx [ae^{ax} \cos bx - be^{ax} \sin bx]$$

$$= e^{2ax} [a \sin^2 bx + b \sin bx \cos bx +$$

$$a \cos^2 bx - b \sin bx \cos bx]$$

$$= ae^{2ax}$$

Sol.66. (a)

$$y = (-\sin x)^{-x}$$

$$\frac{dy}{dx} = (-\sin)^{-x} \left[\frac{x}{\sin x} (-\cos x) + \log(-\sin x) \cdot (-1) \right]$$

$$\left[\frac{dy}{dx} \right]_x = -\frac{\pi}{6} = \left(\frac{1}{2} \right)^{\pi/2} \left[-\frac{\pi}{6} \sqrt{3} - \log \frac{1}{2} \right]$$

$$2^{-\pi/6} \left(\frac{6 \log 2 - \sqrt{3}\pi}{6} \right)$$

Sol.67. (a)

$$\text{For } \frac{\pi}{4} < x < \frac{\pi}{2}, \sqrt{1 - \sin 2x} = \sin x - \cos x$$

$$\therefore \text{Result} = \cos x + \sin x$$

Sol.68. (d)

$$f'(x) =$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)[f(h)-1]}{h} = \lim_{h \rightarrow 0} \frac{f(x)[hg(h)\phi(h)]}{h}$$

$$= abf(x)$$

Sol.69. (a)

$$f(x) = \sec^2(\tan^{-1}x)$$

$$f(x) = 1 + \tan^2(\tan^{-1}x)$$

$$f(x) = 1 + x^2$$

$$f'(x) = 2x$$

Sol.70. (d)

$$f(x) = \sin(\cos x)$$

$$f'(x) = \cos(\cos x) (-\sin x)$$

Sol.71. (a)

$$Y = 2^{(\sin x)^2} \Rightarrow \frac{dY}{dx} = 2^{(\sin x)^2} 2 \sin x \cos x \ln 2$$

$$X = \sin x \Rightarrow \frac{dX}{dx} = \cos x$$

$$\frac{dY}{dX} = 2 \sin x \cdot 2^{(\sin x)^2} \ln 2 = \sin x \cdot 2^{(\sin x)^2} \ln 4$$

Sol.72. (a)

$$x^y = e^{x^y}$$

taking log both sides

$$y \cdot \log x = x - y$$

$$\text{put } x = 1 \text{ then } y = 1$$

differentiate (i) w.r.t. x

$$y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} = 1 - \frac{dy}{dx} \quad \dots(i)$$

put $x = y = 1$ in this equation

$$\frac{dy}{dx} = 0$$

Sol.73. (b)

Diff. equation (ii) of above question

$$x \log y \frac{dy}{dx} - y \frac{d^2 y}{dx^2} + \log x \cdot \frac{d^2 y}{dx^2} + \frac{1}{x} \cdot \frac{dy}{dx} = -\frac{d^2 y}{dx^2}$$

$$\text{put } x = y = 1 \text{ and } \frac{dy}{dx} = 0 \text{ in above equation then}$$

$$\frac{d^2 y}{dx^2} = 1$$

Sol.74. (a)

$$\frac{d(\tan^{-1} x)}{d(\cot^{-1} x)} = \frac{d\left(\frac{\pi}{2} - \cot^{-1} x\right)}{d(\cot^{-1} x)} = -1$$

Sol.75. (b)

$$x^m y^n = a^{m+n}$$

differentiate w.r.t. x

$$mx^{m-1} y^n + ny^{n-1} x^m y' = 0$$

$$x^{m-1} y^{n-1} my + ny^{n-1} x^{m-1} x \cdot y' = 0$$

$$y' = -my/nx$$

Sol.76. (b)

$$e^{\theta \phi} = c + 4\theta \phi$$

diff. w.r.t. θ

$$e^{\theta \phi} \left(\theta \frac{d\phi}{d\theta} + \phi \right) = 4 \left(\theta \frac{d\phi}{d\theta} + \phi \right)$$

$$(e^{\theta \phi} - 4) \left(\theta \frac{d\phi}{d\theta} + \phi \right) = 0$$

$$\theta \frac{d\phi}{d\theta} + \phi = 0$$

$$\theta d\phi = -\phi d\theta$$

Sol.77. (a)

$$Y = e^x$$

$$X = x^e$$

$$\frac{dY}{dX} = e^x$$

$$\frac{dX}{dx} = ex^{e-1}$$

$$\frac{dY}{dX} = \frac{e^x}{ex^{e-1}} = \frac{xe^x}{e^x}$$

Sol.78. (a)

$$y = \sin(\ln x) + \cos(\ln x)$$

$$\frac{dy}{dx} = \cos(\ln x) \cdot \frac{1}{x} - \sin(\ln x) \cdot \frac{1}{x}$$

$$\text{at } x = e$$

$$\frac{dy}{dx} = \cos(\ln e) \cdot \frac{1}{e} - \sin(\ln e) \cdot \frac{1}{e}$$

$$\frac{dy}{dx} = \frac{\cos 1 - \sin 1}{e}$$

Sol.79. (b)

$$x = e^t \cos t$$

$$y = e^t \sin t$$

$$\frac{dx}{dt} = e^t (-\sin t) + e^t \cos t$$

$$\frac{dy}{dt} = e^t (\cos t) + e^t \sin t$$

$$\frac{dx}{dy} = \frac{e^t (-\sin t) + e^t \cos t}{e^t (\cos t) + e^t \sin t}$$

$$\frac{dx}{dy} = \frac{-\sin t + \cos t}{\cos t + \sin t}$$

$$\text{at } t = 0$$

$$\frac{dx}{dy} = 1$$

$$\frac{dy}{dx} = 1$$

Sol.80. (a)

$$Y = e^{e^x} \Rightarrow \frac{dY}{dx} = e^{e^x} \cdot e^x$$

$$X = e^x \Rightarrow \frac{dX}{dx} = e^x$$

$$\frac{dY}{dX} = \frac{e^{e^x} \cdot e^x}{e^x} = e^{e^x}$$

Sol.81. (b)

$$\text{If } y = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$$

$$\text{T.f. } \frac{dy}{dx} \text{ at } x = 0$$

Multiply both sides by $(1-x)$

$$(1-x)y = (1-x)(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$$

$$(1-x)y = (x-x^6)(1+x^{16})$$

$$(1-x)y = 1 - x^{32}$$

$$(1-x) \frac{dy}{dx} = \frac{-32x^{31} + y}{1-x}$$

$$y \text{ at } x = 0 \text{ is } 1$$

$$\frac{dy}{dx} = \frac{-32(3) + (1)}{1} = 1$$

Sol.82. (a)

$$y = \cos x \cdot \cos 4x \cdot \cos 8x \text{ then } \frac{1}{y} \frac{dy}{dx} \text{ at } x = \frac{\pi}{4} \text{ taking}$$

log both sides

$$\log y = \log [\cos x \cdot \cos 4x \cos 8x]$$

$$\log y = \log \cos x + \log \cos 4x + \log \cos 8x$$

$$\frac{1}{y} \frac{dy}{dx}$$

$$= \frac{1}{\cos x} (-\sin x) + \frac{1}{\cos 4x} [-\sin 4x] + \frac{1}{\cos 8x} [-\sin 8x]$$

$$\frac{dy}{dx} = y[-\tan x - 4 \tan 4x - 8 \tan 8x]$$

$$\text{T.f. } \frac{dy}{dx} \text{ at } x = \frac{\pi}{4}$$

$$\text{Here } \frac{1}{y} \frac{dy}{dx} = [-1 - 4 \tan \pi - 8 \tan 2\pi]$$

$$\frac{1}{y} \frac{dy}{dx} = [-1-0-0]$$

$$\frac{1}{y} \frac{dy}{dx} = -1$$

Sol.83. (c)
 $f(x) = x^4$ then $f'(x) = x^2$

$$f'(x) = 2x$$

$$f'(1) = 2$$

Sol.84. (a)
 $x^y \cdot y^x = 1$
 take log both sides
 $y \log x + x \log y = 0$
 diff. w.r.t. x both sides
 $y \cdot \frac{1}{x} + y' \log x + \log y + x \cdot \frac{1}{y} \cdot y' = 0$

$$\text{put } x = y = 1$$

$$1 + 0 + 0 + y' = 0$$

$$y' = -1$$

$$\text{Sol.85. (b)}$$

$$y = (x^x)^x$$

$$y = x^{x^2}$$

$$\log y = x^2 \log x$$

$$\frac{1}{y} \frac{dy}{dx} = x^2 \cdot \frac{1}{x} + 2x \cdot \log x$$

$$\frac{dy}{dx} = y(x + 2x \log x)$$

$$\text{put } x = y = 1$$

$$\frac{dy}{dx} = 1$$

$$\text{Sol.86. (d)}$$

$$y = [x + 1]$$

G.I.f. is constant function between

$$x = -3 \text{ to } x = -4$$

so its derivative is 0

$$\text{Sol.87. (d)}$$

$$f(x) = \frac{m}{x} + 2nx + 1$$

$$f'(x) = \frac{-m}{x^2} + 2n = 0$$

$$= \frac{-m}{4} + 2n = 0$$

$$8n = m$$

$$\Rightarrow m + 8n = m + m = 2m$$

It depends upon m. so can not find due to

sufficient data

$$\text{Sol.88. (c)}$$

$$h(x) = \{f(x)\}^2 + \{g(x)\}^2$$

$$h'(x) = 2f(x) \cdot f'(x) + 2g(x) \cdot g'(x) \dots (i)$$

given that $f'(x) = g(x)$

differentiate both sides

$$f''(x) = g'(x) = -f(x)$$

by equation (i)

$$h'(x) = 2f(x) \cdot g(x) - 2g(x) \cdot f(x) = 0$$

$$\text{so } h'(3) = 0$$

if $h'(x) = 0$, then $h(x)$ is a constant function

$$\text{so } h(1) = h(2)$$

$$\text{Sol.89. (b)}$$

$$y = \ln^2 \left(\frac{x^2 - x + 1}{x^2 + x + 1} \right)$$

$$\frac{dy}{dx} = 2 \ln \left(\frac{x^2 - x + 1}{x^2 + x + 1} \right) \left[\frac{d}{dx} \left(\frac{x^2 - x + 1}{x^2 + x + 1} \right) \right]$$

let differentiation of last step is $g(x)$, then put $x = 0$

$$\frac{dy}{dx} = 2 \ln(1) [g(0)] = 0$$

$$\text{Sol.90. (d)}$$

$$\frac{d}{dx} \left(\frac{1 + x^4 + x^8}{1 - x^2 + x^4} \right)$$

$$= \frac{d}{dx} \left(\frac{(1 + x^2 + x^4)(1 - x^2 + x^4)}{1 - x^2 + x^4} \right)$$

$$= \frac{d}{dx} (1 + x^2 + x^4) = 2x + 4x^3 = ax + bx^3$$

$$a = 2 \text{ and } b = 4$$

$$\text{Sol.91. (c)}$$

$$y = \frac{x\sqrt{x^2 - 16}}{2} - 8 \ln |x + \sqrt{x^2 - 16}|$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2 - 16}}{2} + \frac{x \cdot 2x}{2 \cdot 2\sqrt{x^2 - 16}} - 8 \cdot \frac{1}{x + \sqrt{x^2 - 16}} \left(1 + \frac{2x}{2\sqrt{x^2 - 16}} \right)$$

$$\frac{dy}{dx} = \frac{\sqrt{x^2 - 16}}{2} + \frac{x^2}{2\sqrt{x^2 - 16}} - 8 \left(\frac{1}{\sqrt{x^2 - 16}} \right)$$

solve by taking LCM

$$\frac{dy}{dx} = \frac{x^2 - 16}{\sqrt{x^2 - 16}} = \sqrt{x^2 - 16}$$

$$\text{Sol.92. (a)}$$

differentiation of $\sin^2 x$ w.r.t. $\cos^2 x$

$$\frac{d(\sin^2 x)}{d(\cos^2 x)} = \frac{2 \sin x \cos x}{-2 \sin x \cos x} = -1$$

$$\text{Sol.93. (b)}$$

$$\cos \sec x^\circ = \cos \sec \left(\frac{\pi}{180} x \right)$$

differentiate by chain rule

$$\frac{d}{dx} \cos \sec x^\circ = \frac{d}{dx} \cos \sec \left(\frac{\pi}{180} x \right)$$

$$= -\frac{\pi}{180} \cos \sec \left(\frac{\pi}{180} x \right) \cot \left(\frac{\pi}{180} x \right)$$

$$= -\frac{\pi}{180} \cos \sec(x^\circ) \cot(x^\circ)$$

$$\text{Sol.94. (a)}$$

$$f(x) = |\ln x|$$

at $x = 0.5$, $\ln x$ is negative

$$\text{so } f(x) = -\ln x$$

$$f'(x) = -1/x$$

$$f'(0.5) = -2$$

$$\text{Sol.95. (c)}$$

$$f(x) = |\ln x|$$

at $x = 2$, $\ln x$ is positive

$$\text{so } f(x) = \ln x$$

$$f'(x) = 1/x$$

$$f'(2) = 1/2$$

$$\text{Sol.96. (d)}$$

$$f(x) = |\ln x|$$

for $1 < x < 2$, $f(x)$ is positive but $f(x)$ is

negative

$$\text{so } f(x) = -\ln(x)$$

$$\text{differentiation of } f(x) = -\frac{1}{x \ln x}$$

$$\text{Sol.97. (a)}$$

$$f(x) = |\ln x|$$

at $x = 0.5$, $\ln x$ is negative

$$\text{so } f(x) = -\ln x$$

$$f'(x) = -1/x$$

$$f'(0.5) = -2$$

$$\text{Sol.98. (c)}$$

$$y = f \left(\frac{2x-3}{x} \right) = f \left(2 - \frac{3}{x} \right)$$

$$\frac{dy}{dx} = f' \left(2 - \frac{3}{x} \right) \left(\frac{3}{x^2} \right)$$

given that $f'(x) = \cos(\ln x)$

$$f' \left(2 - \frac{3}{x} \right) = \cos \left(\ln \left(2 - \frac{3}{x} \right) \right)$$

$$\frac{dy}{dx} = \frac{3}{x^2} \cos \left(\ln \left(2 - \frac{3}{x} \right) \right) = \frac{3}{x^2} \cos \left(\ln \left(\frac{2x-3}{x} \right) \right)$$

$$\text{Sol.99. (a)}$$

let

$$g(x) = x - \frac{1}{x} = y$$

$$\text{fog}(x) = x^3 - \frac{1}{x^3}$$

than

$$f(y) = x^3 - \frac{1}{x^3} = \left(x - \frac{1}{x} \right)^3 + 3 \left(x - \frac{1}{x} \right)$$

$$f(y) = y^3 + 3y$$

$$\text{so } f(x) = x^3 + 3x$$

$$g[f(x) - 3x] = g(x^3)$$

$$= x^3 - \frac{1}{x^3}$$

$$\text{Sol.100. (d)}$$

$$f(x) = x^3 + 3x$$

$$f'(x) = 3x^2$$

$$f''(x) = 6x$$

$$\text{Sol.101. (d)}$$

$$h(x) = f(2f(x) + 2)$$

differentiate w.r.t. x

$$h'(x) = f'(2f(x) + 2) \cdot 2f'(x)$$

$$h'(0) = f'(2f(0) + 2) \cdot 2f'(0)$$

$$\text{given that } f(0) = -1$$

$$h'(0) = f'(0) \cdot 2f'(0) = 2$$

$$\text{Sol.102. (a)}$$

$$g(x) = (h(x))^2$$

$$g'(x) = 2h(x) \cdot h'(x)$$

$$g'(0) = 2h(0) \cdot h'(0)$$

by above question $h'(0) = 2$

$$\text{and } h(0) = f(2f(0) + 2) = f(0) = -1$$

$$\text{so } g'(0) = -4$$

$$\text{Sol.103. (c)}$$

$$x = \sec \theta - \cos \theta$$

whole square both side

$$x^2 = \sec^2 \theta + \cos^2 \theta - 2$$

$$x^2 + 4 = \sec^2 \theta + \cos^2 \theta + 2$$

$$x^2 + 4 = (\sec \theta + \cos \theta)^2 \dots (i)$$

$$y = \sec^4 \theta - \cos^4 \theta$$

$$y^2 + 4 = (\sec^4 \theta + \cos^4 \theta)^2 \dots (ii)$$

$$y = \sec^4 \theta - \cos^4 \theta$$

$$\frac{dy}{d\theta} = 4 \sec^3 \theta \cdot \sec \theta \cdot \tan \theta + 4 \cos^3 \theta \sin \theta$$

$$x = \sec \theta - \cos \theta$$

$$\frac{dx}{d\theta} = \sec \theta \cdot \tan \theta + \sin \theta$$

$$\frac{dy}{dx} = \frac{4 \sec^3 \theta \cdot \sec \theta \cdot \tan \theta + 4 \cos^3 \theta \sin \theta}{\sec \theta \cdot \tan \theta + \sin \theta}$$

$$\frac{dy}{dx} = \frac{4 \tan \theta (\sec^4 \theta + \cos^4 \theta)}{\tan \theta (\sec \theta + \cos \theta)}$$

$$\left(\frac{dy}{dx} \right) = \frac{4 (\sec^4 \theta + \cos^4 \theta)}{(\sec \theta + \cos \theta)}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{16 (\sec^4 \theta + \cos^4 \theta)^2}{(\sec \theta + \cos \theta)^2} = \frac{16 (y^2 + 4)}{(x^2 + 4)}$$

$$\text{Sol.104. (c)}$$

By above solution

$$\left(\frac{dy}{dx} \right)^2 = \frac{16 (y^2 + 4)}{(x^2 + 4)}$$

Diff. both sides

$$2 \left(\frac{dy}{dx} \right) \left(\frac{d^2y}{dx^2} \right) = 16 \frac{(x^2+4) \left(2y \frac{dy}{dx} \right) - (y^2+4)(2x)}{(x^2+4)^2}$$

$$(x^2+4)^2 \left(\frac{dy}{dx} \right) \left(\frac{d^2y}{dx^2} \right) = 16(x^2+4) \left(y \frac{dy}{dx} \right) - 16(y^2+4)(x)$$

Divide by $y^2 + 4$

$$\frac{(x^2+4)^2}{(y^2+4)} \left(\frac{dy}{dx} \right) \left(\frac{d^2y}{dx^2} \right) = 16 \frac{(x^2+4)}{(y^2+4)} \left(y \frac{dy}{dx} \right) - 16(x)$$

$$\frac{(x^2+4)^2}{(y^2+4)} \left(\frac{dy}{dx} \right) \left(\frac{d^2y}{dx^2} \right) - 16 \frac{(x^2+4)}{(y^2+4)} \left(y \frac{dy}{dx} \right) = -16x$$

$$\frac{(x^2+4)}{(y^2+4)} \left(\frac{dy}{dx} \right) \left[(x^2+4) \left(\frac{d^2y}{dx^2} \right) - 16y \right] = -16x$$

Sol.105.(d)

$$(x+y)^{p+q} = x^p \cdot y^q$$

Taking log both side

$$(p+q)\log(x+y) = p\log x + q\log y$$

Diff. both sides

$$(p+q) \left(\frac{1}{x+y} \right) \left(1 + \frac{dy}{dx} \right) = \frac{p}{x} + \frac{q}{y} \frac{dy}{dx}$$

$$\left(\frac{dy}{dx} \right) \left(\frac{p+q}{x+y} - \frac{q}{y} \right) = \frac{p}{x} - \frac{p+q}{x+y}$$

$$\left(\frac{dy}{dx} \right) = \frac{\left(\frac{p}{x} - \frac{p+q}{x+y} \right)}{\left(\frac{p+q}{x+y} - \frac{q}{y} \right)}$$

$$\left(\frac{dy}{dx} \right) = \frac{\left(\frac{px+py-px-qx}{x(x+y)} \right)}{\left(\frac{py+qy-qx-qy}{y(x+y)} \right)} = \frac{y(py-qx)}{x(py-qx)} = \frac{y}{x}$$

So derivative is independent of both p and q.

Sol.106.(a)

By above solution $dy/dx = y/x$

Sol.107.(d)

$$y = \sin^{-1} \left(x - \frac{4x^3}{27} \right)$$

$$y = \sin^{-1} \left(3 \frac{x}{3} - 4 \left(\frac{x}{3} \right)^3 \right)$$

Let $x/3 = \sin \theta$

$$y = \sin^{-1} (3 \sin \theta - 4 (\sin \theta)^3) = \sin^{-1} (\sin 3\theta)$$

$$y = 3\theta = 3 \sin^{-1} \left(\frac{x}{3} \right)$$

Sol.108.(c)

By above solution

$$y = 3 \sin^{-1} \left(\frac{x}{3} \right)$$

$$\frac{dy}{dx} = 3 \cdot \frac{1}{\sqrt{1-\frac{x^2}{9}}} \cdot \frac{1}{3} = \frac{3}{\sqrt{9-x^2}}$$

Sol.109.(a)

$$y = f(x) = \frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \ln \sqrt{1-x^2}$$

Put $x = \sin \theta$

$$dx/d\theta = \cos \theta$$

$$f(\sin \theta) = \frac{\sin \theta \sin^{-1}(\sin \theta)}{\sqrt{1-\sin^2 \theta}} + \ln \sqrt{1-\sin^2 \theta}$$

$$y = f(\sin \theta) = \theta \tan \theta + \ln \cos \theta$$

Diff. w.r.t. x

$$\frac{dy}{dx} = (\tan \theta + \theta \sec^2 \theta - \tan \theta) \frac{d\theta}{dx}$$

$$\frac{dy}{dx} = \theta \sec^2 \theta \frac{d\theta}{dx} = \theta \sec^2 \theta \frac{1}{\cos \theta} = \theta \sec^3 \theta$$

Put $x = 1/2$ then $\theta = 30^\circ$

$$f'(0.5) = \frac{\pi}{6} \left(\frac{2}{\sqrt{3}} \right)^3 = \frac{4\sqrt{3}\pi}{27}$$

Sol.110.(c)

$$\frac{dy}{dx} = \theta \sec^3 \theta$$

$$\frac{d^2y}{dx^2} = (\theta \cdot 3 \sec^2 \theta \sec \theta \tan \theta + \sec^3 \theta) \frac{1}{\cos \theta}$$

At $x = 0$, $\theta = 0^\circ$

$$\frac{d^2y}{dx^2} = (0+1) \frac{1}{1} = 1$$

Sol.111.(c)

By above solution

$$\frac{dy}{dx} = \theta \sec^3 \theta$$

Sol.112.(c)

For $x = -1$

$$f(x) = 4(5^x)$$

$$f'(x) = 4(5^x \log 5)$$

$$f'(-1) = 4(5^{-1} \log 5) = \frac{4}{5} \ln 5$$

Sol.113.(a)

$$f(x) = |x-3| + |x-4|$$

for $3 < x < 4$

$$f(x) = (x-3) - (x-4) = 1$$

$$dy/dx = 0$$